



Ride comfort evaluation of stochastic traffic flow crossing long-span suspension bridge experiencing vortex-induced vibration

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ABSTRACT

The long-span bridge is susceptible to various wind-induced vibrations owing to its high flexibility and low structural damping, among which the vortex-induced vibration (VIV) frequently emerges in recent years resulting in considerable public concerns. Although the VIV is a limited cyclic vibration, it can cause significant negative effects on the ride comfort of vehicles crossing the bridge. Evaluating the ride comfort of vehicles crossing long-span bridges under VIV, therefore, becomes urgent to make decisions on continuing or closing the operations of bridges. In view of this, this paper presents a general ride comfort evaluation procedure for stochastic traffic flow crossing the long-span bridge under VIV, which adequately considers the realistic traffic behavior and the complex interactions among wind, vehicles (traffic flow), and bridge. The developed procedure is validated through the field-monitoring data collected in a recent VIV event of a prototype long-span suspension bridge. On this basis, the ride comfort of vehicles in different traffic flows crossing the prototype bridge under VIV is evaluated through the criteria recommended in the ISO 2631-1 standard based on the overall vibration total value (OVTV). Moreover, the effects of traffic density, traffic proportion, and road roughness on drivers' ride comfort are studied.

1. Introduction

With the ongoing development of regional integration and economic globalization, there is a high demand for efficient traffic networks that reduce factor costs through shortening the spatial-temporal distances among various regions. As the key node of the sizeable transportation infrastructures, such as high-speed railways and sea-crossing channels, a growing number of long-span bridges have been constructed in many countries worldwide (Zhu and Zhang, 2017). One of the major challenges confronted by these super long bridges is the wind-induced vibration problem, due to the increasing streamlining of the main beam section. Among a variety of wind-induced vibrations, the vortex-induced vibration (VIV) is the most common one, which occurs easily on long-span bridges under relatively low wind speed. Although VIV is not as destructive as the flutter, its frequent occurrence accompanied with limited vibration amplitude could result in fatigue damage accumulation in bridge components and escalate the comfort levels of vehicle users or pedestrians.

VIV has been occasionally observed on several long-span bridges

worldwide. For example, in 1980, the slender twin-box-girder of Rio-Niteroi Bridge in Brazil exhibited occasional vortex-induced vertical oscillations in the wind speed range of 15.3 m/s to 16.7 m/s (Battista and Pfeil, 2000). The Tokyo Bay Bridge in Japan incurred VIV in the first vertical bending mode with a maximum amplitude up to 0.5m, under a mean wind speed of 17 m/s that is nearly normal to the bridge axis, in February 1995 (Yang et al., 2015). The Second Severn Crossing cable-stayed bridge in Great Britain suffered from strong VIVs in the first winter after it was opened to traffic in 1996 (Macdonald et al., 2002). Other examples of VIVs on actual bridges include the Great Belt East suspension bridge in Denmark (Frandsen, 2001), the Volgograd continuous bridge with steel box girder in Russia (Weber and Maslanka, 2012), the Xihoumen Bridge in China (Li et al., 2011), and the Yi Sun-sin suspension bridge in South Korea (Hwang et al., 2015). Most recently, large-amplitude VIVs were observed on three suspension bridges in China including the Xihoumen suspension bridge, Humen suspension bridge and Yingwuzhou Yangtze River Bridge during April to May 2020 (Zhao et al., 2020). The surveys reported that some of the drivers and passengers who were on the above three bridges under VIVs experienced

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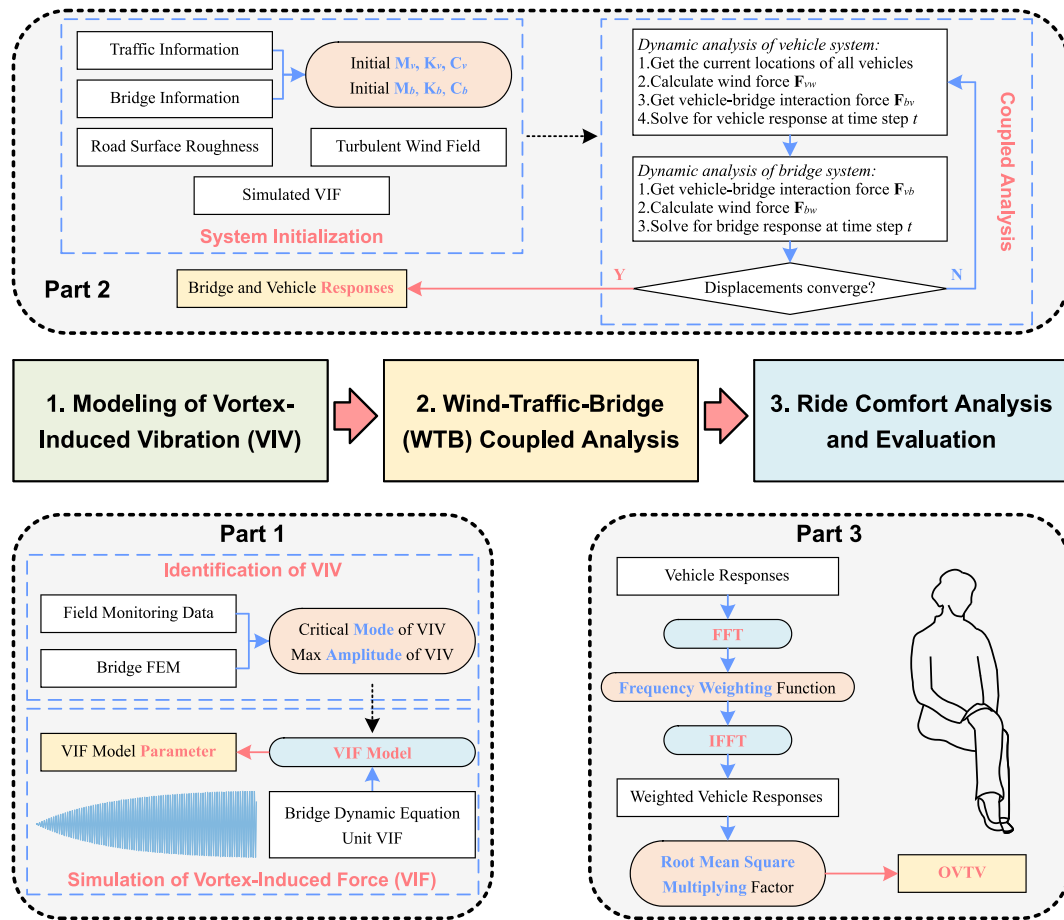


Fig. 1. General procedure of ride comfort evaluation.

various degrees of short-term body discomfort, although it only took a few minutes for the vehicles running through the bridges (Zhao et al., 2020).

As discussed previously, although the VIV is not a divergent vibration, it may still cause appreciable vehicle dynamic vibrations, resulting in driver's ride discomfort and bringing up concerns of the passengers, which in turn reduces the serviceability of long-span bridges. This implies the necessity of conducting the driver's ride comfort evaluation on long-span bridges under VIV, to reduce the risk of driver's discomfort and improve the serviceability of bridges. Nevertheless, most studies in literature typically focused on the driver's ride comfort in rigid roadways under normal wind conditions, e.g., Nahvi et al. (2009). Very few studies have considered the driver's ride comfort running on long-span bridges under crosswind, as this entails a complex wind-vehicle-bridge (WVB) problem that can only be tackled by adequately quantifying the coupling between the bridge dynamic response and the moving vehicles. Typically, the drivers' ride comfort is studied by evaluating the human exposure to whole-body vibration mainly based on the specifications BS 1987 (BSI, 1999) and ISO 2631-1 (ISO, 1997). The whole-body vibration referred to the vehicle vibrations transmitted to the occupational drivers in various directions through their contact surfaces including the back, seat, and footrest. Xu and Guo (2004) investigated the ride comfort of high-sided vehicles crossing a cable-stayed bridge and pointed out that the crosswind and the bridge motion controlled the ride comfort in the lateral and vertical directions, respectively. This was echoed by Nguyen et al. (2017), who also concluded that the wind-induced bridge lateral vibration could cause potential discomfort and fatigue of the drivers running on a very slender arch bridge based on coupled WVB analysis. Yin and his colleagues (Yin et al., 2011) analyzed the ride comfort of vehicles moving through a

high-pier multi-span continuous bridge using the procedures recommended in ISO 2631-1, in which the drivers' comforts were found to be affected by both the bridge lateral and longitudinal vibrations. Later on, Zhu et al. (2018a) evaluated the drivers' ride comfort when vehicles were travelling on a coastal cable-stayed bridge under correlated wind and wave loads via the overall vibration total value (OVTV) as recommended in the ISO 2631-1 standard. The OVTV is computed using frequency-weighted RMS values based on multi-axis whole-body vibrations, which is subsequently utilized to determine the drivers' ride comfort level. Most recently, Camara et al. (2019) examined the drivers' ride comfort considering the transverse correlation of the pavement irregularities and wind actions. The results revealed that the drivers' ride comfort was mostly influenced by both the random pavement irregularities and crosswind.

Despite the above endeavors, the simultaneous presence of multiple vehicles on the bridge (i.e., traffic flow) is seldom addressed when performing the WVB analysis. It is not until Chen and Wu, 2010, 2011 introduced the microscopic cellular automaton (CA) traffic model into bridge analysis that people started to consider the real-world vehicle behavior in the Traffic-Bridge (TB) and Wind-Traffic-Bridge (WTB) analysis, such as acceleration, deceleration, and lane-changing. Several studies (Liu et al., 2016; Yin et al., 2016; Zhou and Chen, 2018) have shown that long-span bridges exhibit distinct dynamic behaviors under CA-based stochastic traffic flow, which is more in accordance with the realistic traffic loading scenario. On this basis, Zhou and Chen (2016) evaluated the ride comfort of three different vehicle types travelling on a cable-stayed bridge accounting for complex dynamic interactions among bridge, realistic traffic, and wind actions. An appreciable influence of random traffic and wind loads on the drivers' ride comfort was observed. Moreover, Liu et al. (2016) highlighted the importance of

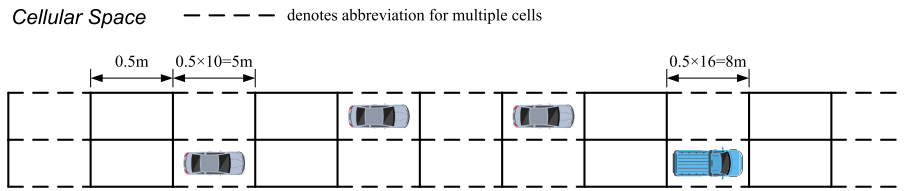


Fig. 2. Schematic diagram of the LAI-E model.

incorporating the stochastic traffic follow and bridge surface progressive deterioration into the drivers' ride comfort. Nonetheless, the commonly used CA traffic model in TB/WTB analysis proposed by Nagel and Schreckenberg (NaSch model) has proved oversimplified in the real-world traffic pattern reproduction regarding vehicular dimension, acceleration/deceleration capability, and movement (Hou et al., 2017; Lan et al., 2009; Lárraga and Alvarez-Icaza, 2010, 2014; Li et al., 2016), which may inevitably detract the accuracy of corresponding TB/WTB analysis and ride comfort evaluation. It is noted that all works mentioned above considered only the normal wind condition, while the adverse wind condition, such as VIV, remains challenging to be considered.

To address the aforementioned limitations, the present study aims to develop a general ride comfort evaluation procedure for the stochastic traffic flow crossing the slender long-span bridge experiencing VIV, which contains three steps: (1) modelling the VIV of the long-span bridge through a vortex-induced force (VIF) model; (2) conducting WTB coupled analysis with considering realistic traffic behavior; (3) performing ride comfort evaluation as recommended by ISO 2631-1. Then, a field-monitoring VIV case of a long-span suspension bridge is introduced for validation and demonstration. Finally, numerical analyses are performed to identify key factors (e.g., traffic density, traffic proportion, and road roughness) that may affect the response and ride comfort of stochastic traffic flow crossing a long-span suspension bridge experiencing VIV. To the authors' best knowledge, it is the first attempt pertaining to understand the ride comfort of vehicles in realistic traffic flow on the long-span bridge under VIV through integration of numerical simulation and field-monitoring data.

2. Ride comfort evaluation procedure

The ride comfort evaluation procedure for the stochastic traffic flow crossing a long-span bridge experiencing VIV is built upon preliminary works of the authors (Zhu and Zhang, 2017; Zhu et al., 2018a, 2018b; Xiong et al., 2020), which is composed of three parts, i.e., modelling of VIV (part I), WTB coupled analysis (part II) and ride comfort analysis and evaluation (part III). Fig. 1 illustrates the whole evaluation procedure, which will be introduced in detail in the following subsections.

2.1. Modelling of vortex-induced vibration

As the fundamental part of the ride comfort evaluation procedure, it is crucial to develop a suitable approach to simulate the VIV of the bridge, where one intuitive way is to apply the vortex-induced force (VIF) to the finite element model (FEM) of the bridge. However, since bridge decks are generally elastic bluff bodies, it is currently still quite challenging to establish reliable mathematical models of VIF on bridge decks via pure theoretical ways based on the basic flow principles (Zhu et al., 2013). To overcome this difficulty, the empirical VIF model, building upon the mathematical models with partial parameters determined from the wind tunnel tests, becomes an alternative approach.

During the past several decades, various empirical models of uncoupled VIF via wind tunnel tests have been proposed, including single-degree-of-freedom models (Larsen, 1995; Simiu and Scanlan, 1996; Zhu et al., 2017) and 2-DOF models considering both the structural and flow oscillators (Diana et al., 2006; Xu et al., 2015). These

empirical models provide valuable insights into the mechanism of vortex shedding from bluff bridge decks. Nevertheless, so far, there is a lack of a model that is universally better for all practical engineering applications, because the aerodynamic property of a bridge deck varies remarkably with the variations of the deck forms and shapes. This indicates there exists a relatively suitable VIF model for a bridge deck of a specific form and shape, but the search for such a model may require extensive wind tunnel tests. As an initial step, the present study starts from the simple harmonic VIF model proposed by Ruscheweyh and Sedlacek (1988), which only considers the pure vortex-induced force due to the shedding of vortices from the bluff body. It should be noted that a more suitable VIF model considering both the pure vortex-induced force component and motion-induced force component can be easily included into the developed procedure once it is determined through wind tunnel test in the future. The pure vortex-induced force acting on the per-unit length deck can be expressed as (Ruscheweyh and Sedlacek, 1988):

$$F_{VI} = \frac{1}{2} \rho U^2 D \tilde{C}_L \sin(\omega t + \theta) = a \sin(\omega t + \theta) \quad (1)$$

where ρ denotes the air density; U denotes the wind speed; D denotes the deck depth; $\tilde{C}_L$ denotes the root-mean-square value of the lift aerodynamic force coefficient; a denotes the product of these parameters; ω denotes the circular frequency of the vortex shedding, which can be computed as $\omega = 2\pi f_b$ and f_b is the frequency of the VIV; t denotes the time in seconds; θ denotes the phase difference between the VIF and the VIV. Traditionally, the parameters in Eq. (1) can be determined through massive wind tunnel tests for a specific kind/mode of bridge VIV. However, it is prohibitive to conduct such wind tunnel tests for all long-span bridges of interest. Meanwhile, owing to the uncertain nature of wind-induced vibration, the VIV occurring in actuality may vary from the experiment results. Therefore, an alternative way proposed in this work is to identify the parameter of the VIF model through field-monitoring data and bridge FEM, and the detailed process will be illustrated in subsection 3.1.3 with a demonstration case.

2.2. Wind-traffic-bridge coupled analysis

To rationally assess the complex interactions between wind, multiple vehicles (traffic), and bridge, a framework of WTB coupled analysis considering realistic traffic behavior recently proposed by the authors (Xiong et al., 2020) is introduced in the present study, which provides a fundamental basis for the ride comfort evaluation. For brevity, this subsection only provides a succinct introduction of the WTB coupled analysis platform developed in previous works of the authors. Interested readers may refer to the literatures (Zhu et al., 2018a, 2018b; Xiong et al., 2020) for more details.

2.2.1. Stochastic traffic flow simulation

Herein, the stochastic traffic flow running on the long-span bridge is simulated based upon an improved CA model (i.e., LAI-E model) to resolve the limitations of NaSch model as mentioned above. As shown in Fig. 2, it is assumed in LAI-E model that both time and space are discrete, and each lane is divided into cells with an identical length (e.g., 0.5 m). For example, the sedan car with a length of 5 m can occupy 10 cells simultaneously. Meanwhile, each cell in the LAI-E traffic model can be

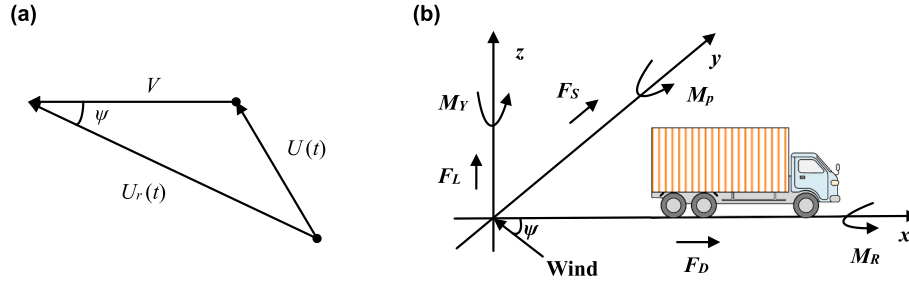


Fig. 3. A schematic diagram of the wind loads acting on the driving vehicle: (a) relative wind velocity; (b) wind load vector.

empty or occupied by up to one vehicle, and each vehicle can occupy several cells simultaneously according to its actual length. Furthermore, the uniformly accelerating/decelerating movement assumption is introduced to represent the kinetics theory. On this basis, the position and velocity of each vehicle will be updated parallelly at each time step via single-lane forwarding and lane-changing rule, which can be found in (Xiong et al., 2020).

2.2.2. Finite element modelling of vehicle and bridge

Once the running trajectory of the vehicle is generated through LAI-E model, each vehicle is further modelled as a combination of several rigid bodies and wheel axles connected by springs and dampers. The mass matrices $\mathbf{M}_v$, stiffness matrices $\mathbf{K}_v$, and damping matrices $\mathbf{C}_v$ of the vehicle can be obtained from the vehicle's equations of motion according to its own properties (Cai and Chen, 2004).

In the present study, the long-span suspension bridge is modelled as a three-dimensional (3D) finite element model using three types of elements, i.e., the 3D beam element for the girder and pylon, the 3D link element for the cable and suspender, and the mass-only element to consider the masses of pavement and railing. On this basis, the mass matrices $\mathbf{M}_b$ and the stiffness matrices $\mathbf{K}_b$ of the bridge can be extracted. Moreover, the Rayleigh damping assumption is adopted to construct the bridge damping matrices $\mathbf{C}_b$.

2.2.3. Dynamic excitation simulation

The dynamic excitations of the WTB system originate from two parts: (1) road roughness of the bridge; (2) turbulent wind field, which will be introduced briefly in this subsection.

According to ISO 8608 specification (ISO, 1995), the road surface roughness profile is assumed as a stationary Gaussian random process, which can be generated by applying the inverse fast Fourier transformation (IFFT) (Huang and Wang, 1992):

$$r(x) = \sum_{i=1}^N \sqrt{4S(f_i)\Delta f} \cos(2\pi f_i x + \theta_i) \quad (2)$$

$$S(f) = S(f_0)(f/f_0)^{-\alpha} \quad (3)$$

where S is the power spectral density (PSD) function of $r(x)$; f_0 is the discontinuity circular frequency; $S(f)$ is the roughness coefficient; $\alpha = 2$ is an exponent of the PSD; f is the spatial frequency; $\Delta f = 1/Nx_0$; $x_0 = 0.02$ m is the distance interval between the successive ordinates of the surface profile; $N = 2048$ is the number of data points; and θ_i is a set of independent random phase angles uniformly distributed in $[0, 2\pi]$. ISO 8608 (ISO, 1995) classifies the road roughness levels in terms of the roughness coefficients, and the road roughness are classified as class A to class E, representing the very good, good, average, poor, and very poor road surface, respectively.

The turbulent wind field is usually decomposed into a mean wind speed and three turbulent wind components for simple implementation in numerical simulations. Generally, the turbulent wind components are treated as stationary Gaussian stochastic processes that can be generated

through the spectral representation method. The wind forces on the vehicle and the bridge are further obtained from their aerodynamic characteristics based on the simulated wind field (Zhu et al., 2018a, 2018b). In the present study, as the wind force on the bridge is modelled as VIF through the VIF model and field-monitoring data, the turbulent wind field is mainly generated to calculate the wind force on the vehicle, which will be illustrated subsequently.

2.2.4. Governing equation

The governing equations of motion for the WTB coupled dynamic system are as follows:

$$\mathbf{M}_b \ddot{\mathbf{u}}_b + \mathbf{C}_b \dot{\mathbf{u}}_b + \mathbf{K}_b \mathbf{u}_b = \sum_{i=1}^n \mathbf{F}_{vb}^i + \mathbf{F}_{bw} \quad (4)$$

$$\mathbf{M}_v^i \ddot{\mathbf{u}}_v^i + \mathbf{C}_v^i \dot{\mathbf{u}}_v^i + \mathbf{K}_v^i \mathbf{u}_v^i = \mathbf{F}_{vG}^i + \mathbf{F}_{vw}^i + \mathbf{F}_{bv}^i \quad (5)$$

where the superscript i ($i = 1, 2, \dots, n$) denotes i th vehicle on the bridge; the subscripts b and v denote the bridge and vehicle; $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$ are the mass, damping, and stiffness matrices, respectively; $\mathbf{u}$, $\dot{\mathbf{u}}$, $\ddot{\mathbf{u}}$ are the nodal displacement, velocity, and acceleration vectors, respectively. The force term $\mathbf{F}_{vG}^i$ on the right-hand side of Eq. (5) denotes the weight of the i th vehicle. The other force terms on the right-hand side of Eqs. (4) and (5) are introduced briefly as follows.

The force term $\mathbf{F}_{bv}^i$ and $\mathbf{F}_{vb}^i$ are the coupled vehicle-bridge forces, i.e., action and reaction forces existing at the contact points between i th vehicle and the bridge, which are function of deformation of the vehicle's lower spring (Cai and Chen, 2004):

$$\mathbf{F}_{bv}^i = -\mathbf{F}_{vb}^i = \mathbf{K}_l \{Z_a - Z_b - r(x)\} + \mathbf{C}_l \left\{ \dot{Z}_a - \dot{Z}_b - \dot{r}(x) \right\} \quad (6)$$

where $\mathbf{K}_l$ and $\mathbf{C}_l$ are the coefficients of the vehicle's lower spring and damper; Z_a is the vehicle-axle-suspension displacement; Z_b is the displacement of the bridge at road-tire contact points; $r(x)$ is the road surface profile; and $\dot{r}(x) = (dr(x)/dx)(dx/dt) = (dr(x)/dx)V(t)$ and $V(t)$ is the vehicle velocity.

The $\mathbf{F}_{bw}$ in Eq. (4) denotes the vertical vortex-induced force vector acting on the bridge deck, which can be expressed as:

$$\mathbf{F}_{bw} = \int_0^L \varphi_n(x) \cdot F_{Vl}(t) dx \quad (7)$$

where $\varphi_n(x)$ denotes the n th mode shape of the bridge (i.e., the mode shape of VIV); L denotes the length of bridge span; F_{Vl} denotes the vortex-induced force (VIF) on bridge deck per unit length which is determined using Eq. (1).

The force term $\mathbf{F}_{vw}^i$ in Eq. (5) denotes the aerodynamic force vector acting on the driving vehicle, which consists of six components, i.e., side force F_S , lift force F_L , drag force F_D , pitching moment M_P , yawing moment M_Y , and rolling moment M_R , as shown in Fig. 3. The $\mathbf{F}_{vw}^i$ can be expressed as (Baker, 1986):

$$\begin{cases} F_S = 0.5\rho U_r^2(t)C_S(\psi)A_0; F_L = 0.5\rho U_r^2(t)C_L(\psi)A_0; F_D = 0.5\rho U_r^2(t)C_D(\psi)A_0 \\ M_P = 0.5\rho U_r^2(t)C_P(\psi)A_0h_v; M_Y = 0.5\rho U_r^2(t)C_Y(\psi)A_0h_v; M_R = 0.5\rho U_r^2(t)C_R(\psi)A_0h_v \end{cases} \quad (8)$$

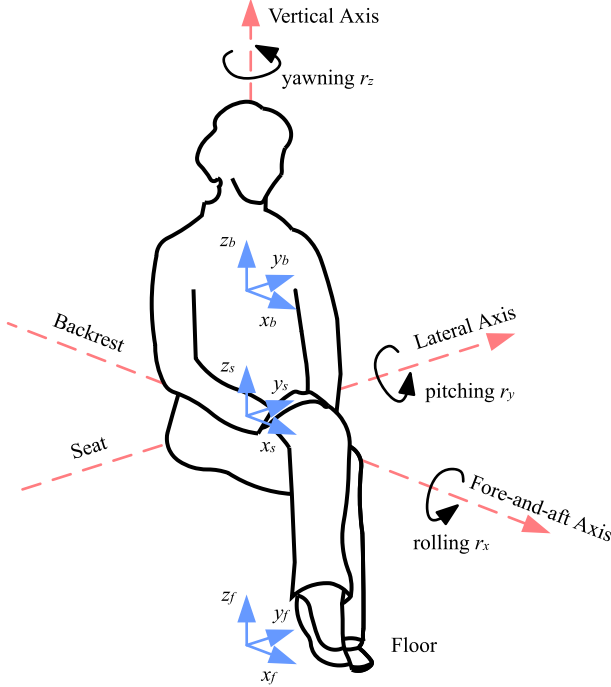


Fig. 4. Vibration measure for a seated person.

where $C_S(\psi)$, $C_L(\psi)$, $C_D(\psi)$, $C_P(\psi)$, $C_Y(\psi)$ and $C_R(\psi)$ denote the wind aerodynamic coefficients that are functions of yaw angle ψ , and ψ is defined as the angle between the direction of the incoming wind and the vehicle driving direction (see Fig. 3(a)); A_0 is the frontal area of the vehicle; h_v is the distance from the vehicle gravity center to the road surface; U_r is the relative wind velocity to the vehicle which is obtained, as shown in Fig. 3(b), from the vector subtraction of vehicle driving speed, V , from the wind speed, U .

Noteworthy, the mutual aerodynamic interference effect between vehicles is beyond the scope of the present study and is not considered herein. Although some preliminary studies have proved that the aerodynamic coefficients of multiple vehicles running on the bridge would change after considering the mutual aerodynamic interference effect (Han et al., 2019; He et al., 2020), it is still very challenging to evaluate the mutual aerodynamic interference effect in the transient WTB coupled analysis. Therefore, the mutual aerodynamic interference effect deserves further endeavors, and it can be incorporated into the existing framework once rationally modelled.

2.3. Ride comfort analysis and evaluation

It is very challenging to evaluate the human tolerance to vehicle vibration because the human vibration tolerance can be affected by many factors. Therefore, there is no consensus on the criterion of ride comfort evaluation. In this work, the widely used whole-body vibration measure recommended by ISO 2631-1 (ISO, 1997) is adopted for the drivers' ride comfort evaluation, which considers all the motions transmitted to the human body through all supporting surfaces. For a seated person under investigation, there are three supporting surfaces, i. e., the seat, backrest, and floor, as shown in Fig. 4. Meanwhile, Fig. 4

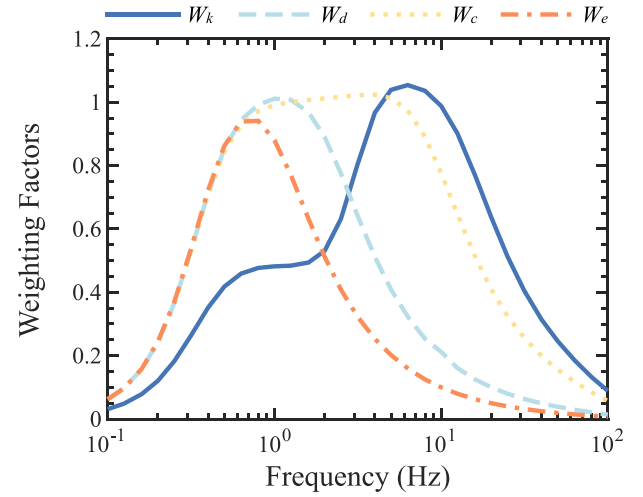


Fig. 5. Weighting curve for different participating axes.

also illustrates the 12 axes where the vibration is measured for a seated person. In the context of WTB system, the vehicle response at the seat, backrest, and floor surface of the driver can be obtained through coupled analysis. Since the fore-and-aft and yawing accelerations are insignificant in this context (Zhu et al., 2018a; Zhou and Chen, 2016), a total of 8 types of vibration are considered for ride comfort evaluation: vertical and lateral vibrations at all three supporting surfaces; pitching and rolling vibrations at the seat location. The eight vibrations can be obtained from the vehicle centroid acceleration response calculated by WTB coupled analysis, considering their relative positions to the vehicle centroid as follows:

$$a_{vs} = a_{vb} = a_{vf} = \ddot{Z}_v \quad (9a)$$

$$a_{ls} = a_{lb} = a_{lf} = \ddot{Y}_v \quad (9b)$$

$$a_{ps} = \ddot{\theta}_v \cdot d_s \quad (9c)$$

$$a_{rs} = \ddot{\phi}_v \cdot y_s + \frac{1}{2} \ddot{\phi}_v \cdot h_s \quad (9d)$$

where a_{ij} denotes the vehicle acceleration: the subscript i denotes the response direction where v , l , p , and r represent vertical, lateral, pitching, and rolling direction; the subscript j denotes the response axis location where s , b , and f represent seat, backrest, and floor; $\ddot{Z}_v$, $\ddot{Y}_v$, $\ddot{\theta}_v$,

Table 1
Multiplying factors for each vibration measure.

Multiplying Factor	Value	Location	Direction
M_k	1.00	Seat	Vertical
M_d	1.00	Seat	Lateral
M_e	0.40	Seat	Pitching
M_e	0.20	Seat	Rolling
M_c	0.40	Backrest	Vertical
M_d	0.50	Backrest	Lateral
M_k	0.40	Floor	Vertical
M_k	0.25	Floor	Lateral

and $\ddot{q}_v$, represent vertical, lateral, pitching, and rolling accelerations at the vehicle centroid; d_s , y_s , and h_s are longitudinal, transverse, and vertical distances between the vehicle centroid and the driver's seat.

According to ISO 2631-1 (ISO, 1997), the calculated accelerations should be further frequency-weighted to model the influence of acceleration responses with different frequencies on the human body. In the present study, a fast Fourier transform (FFT) convolution is adopted.

$$\text{OVTV} = \sqrt{M_k^2(\text{RMS}_{v_s}^2 + \text{RMS}_{v_f}^2 + \text{RMS}_{ij}^2) + M_d^2(\text{RMS}_{l_s}^2 + \text{RMS}_{l_b}^2) + M_e^2(\text{RMS}_{p_s}^2 + \text{RMS}_{p_b}^2) + M_c^2 \text{RMS}_{v_b}^2} \quad (11)$$

Specifically, the time history $x(k)$ (with N discrete values) of the original acceleration is transformed into frequency domain $X(r)$ using the discrete Fourier transform (DFT) as follow:

$$X(r) = \sum_{k=0}^{N-1} x(k) \omega_N^{rk} \quad (10)$$

where $\omega_N = e^{-2\pi i/N}$, $r = 0, 1, \dots, N-1$. Then, $X(r)$ is multiplied by the corresponding frequency weighting factor (Fig. 5), in which W_k , W_d , W_e , and W_c are related to vertical vibration at the seat and vertical and lateral vibrations at the floor, lateral vibration at the seat and floor, rotational vibration at the seat, and vertical vibration at the backrest, respectively. Later, the weighted acceleration response in the frequency domain $X'(r)$ is transformed back into the time domain as $x'(k)$ through the inverse DFT.

The ISO 2631-1 (ISO, 1997) recommends when the ride comfort is

Table 2
Comfort criteria based on OVTV value.

OVTV value(m/s ²)	Comfort Level
<0.315	Not Uncomfortable (Level 1)
0.315–0.630	A Little Uncomfortable (Level 2)
0.500–1.000	Fairly Uncomfortable (Level 3)
0.800–1.600	Uncomfortable (Level 4)
1.250–2.500	Very Uncomfortable (Level 5)
>2.000	Extremely Uncomfortable (Level 6)

affected by vibrations in more than one direction/axis (i.e., the case for a seated person in this study), the overall vibration total value (OVTV) should be adopted for ride comfort evaluation. In specific, the OVTV is obtained from the root-mean-square (RMS) values of the weighted acceleration response with multiplying factors (Table 1) as shown in Eq. (11), which serves to compensate the varying vibration effects from different locations and directions:

in which M_k , M_d , M_e , and M_c are multiplying factors for vertical vibration at the seat and vertical and lateral vibrations at the floor, lateral vibration at the seat and floor, rotational vibration at the seat, and vertical vibration at the backrest, respectively; RMS_{ij} denotes the RMS value of the frequency-weighted acceleration, and the subscripts “i” and “j” share the same definitions as vehicle acceleration in Eq. (9).

After the OVTV is calculated through Eq. (11), the drivers' ride comfort is assessed according to the criteria of different discomfort levels in Table 2 as recommended by ISO 2631-1 (ISO, 1997).

3. Demonstration case

3.1. Prototype bridge

3.1.1. Synopsis of Yingwuzhou Yangtze River Bridge

The Yingwuzhou Yangtze River Bridge (YWB), opened to road traffic in December 2014, is a three-tower four-span suspension bridge carrying the southern section of the Second Ring Road over the Yangtze River in Wuhan, Hubei Province, China. The YWB has a total length of 2100 m, with a span arrangement of 200 m + 2 × 850 m + 200 m, as shown in Fig. 6(a). The span length of the main cables, separated by a transverse distance of 36 m, over the two spans of the bridge is 850 m with a sag-to-span ratio of 1/9, while the span length of the main cables over the side spans is 225 m. The three towers are not in equal height, of which the intermediate steel-concrete hybrid tower is 152 m high, and the two side concrete towers are 126.2 m high. All three towers are constructed on

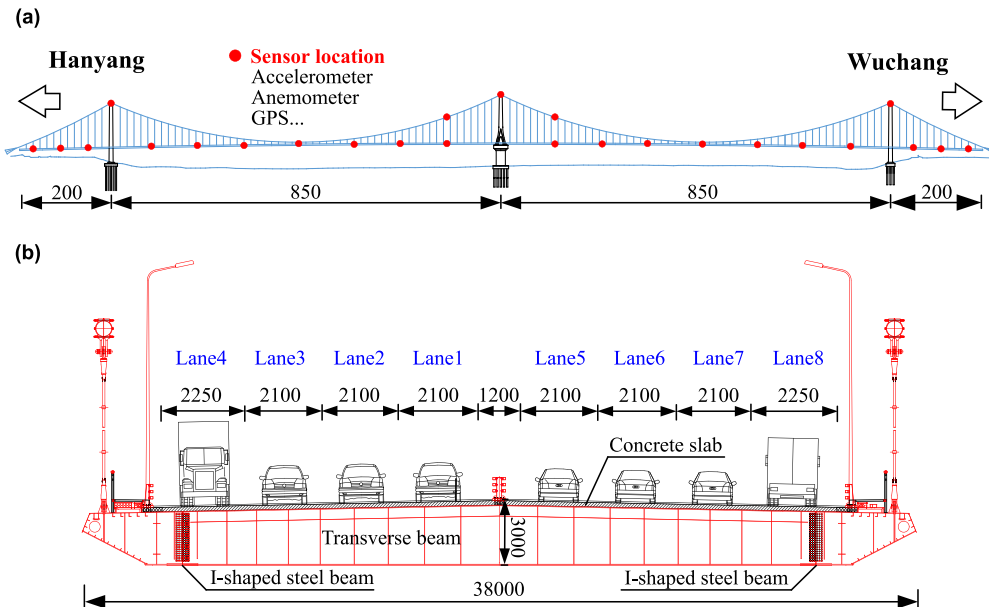


Fig. 6. Yingwuzhou Yangtze River Bridge: (a) elevation view of the bridge (m); (b) cross section of main girder (mm).

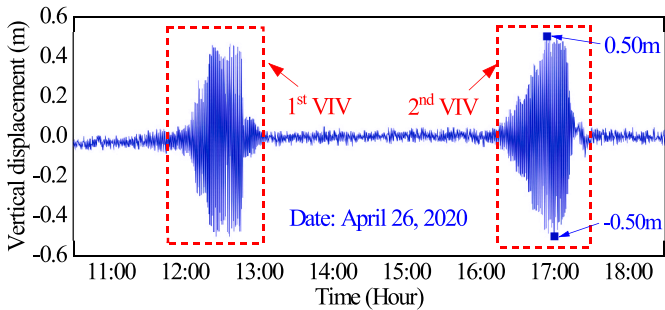


Fig. 7. Measured vertical displacement time history at 3/8 span of Hanyang-side main span.

piled foundations. The π -shaped steel-concrete composite stiffening girder is adopted for the bridge span, which has a cross section that is 38 m in width and 3 m in height, as shown in Fig. 6(b). The concrete slab of the girder has a thickness of 0.2 m and is connected with two

longitudinal I-shaped steel beams with 31.2 m apart. The transverse beam is placed along the bridge longitudinal axis with an equal distance of 3 m. The bridge span carries two-way road traffic, each way having four lanes as illustrated in Fig. 6(b).

3.1.2. Observed vortex-induced vibration

To better acquire the performance of YWB, a structural health monitoring system (SHM) was installed on the bridge, which consists of various types of sensors, including anemometers, accelerometers, displacement gauges, GPSs, strain gauges, thermistors, etc. With the SHM system, two vortex-induced vibration (VIV) events on April 26, 2020, were observed and recorded, one lasting for a duration of 80 min (11:40–13:00) and the other lasting for a duration of 70 min (16:20–17:30). Though the time interval between these two VIV events is about 3 h, the data collected by SHM revealed that the wind condition and the dynamic characteristics of the bridge were very similar, which is briefly introduced herein. During both VIV events, the wind velocity maintained 7.0–7.6 m/s at the deck level with a yaw angle of approximately 80°. The wind yaw angle is defined as the angle between the

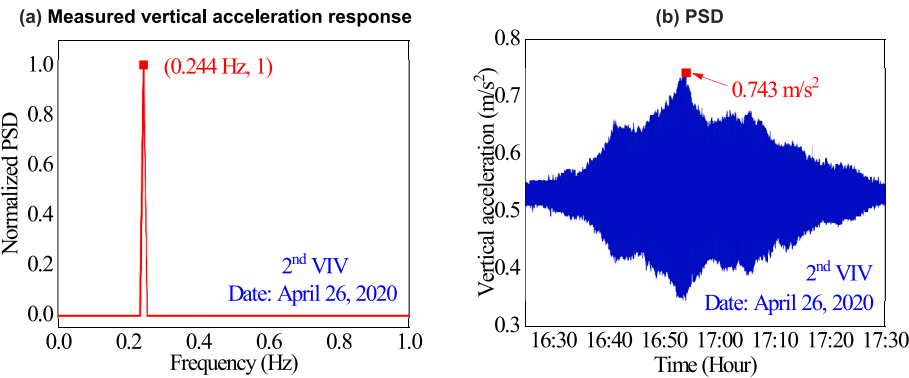


Fig. 8. Measured vertical acceleration response at 3/8 span of Hanyang-side main span during 2nd VIV event and the corresponding PSD. (a) Measured vertical acceleration response (b) PSD.

Table 3
First eight vertical modes of YWB.

Vertical Mode	Natural Frequency (Hz)	Mode Shape Description
VAS1	0.101	
VS1	0.128	
VAS2	0.128	
VS2	0.161	
VAS3	0.183	
VS3	0.211	
VAS4	0.242	
VS4	0.289	

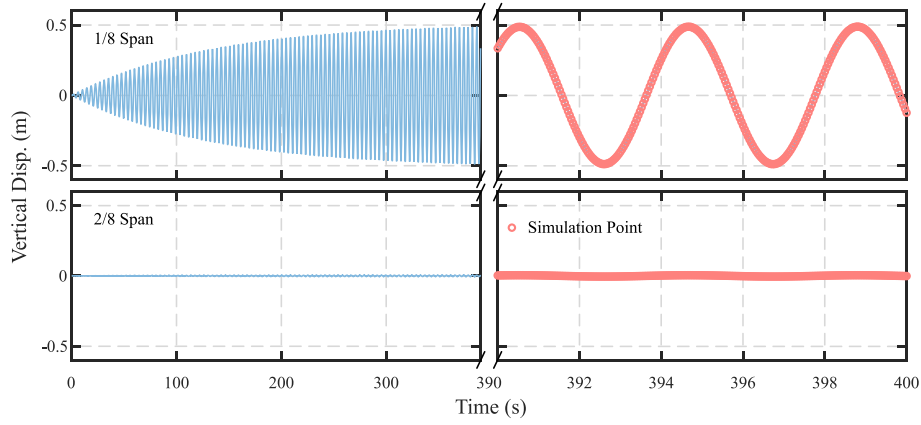


Fig. 9. Vertical bridge displacement over time under simulated VIV.

direction of the incoming wind and the bridge longitudinal axis, and a yaw angle of 80° indicates that the wind direction is nearly perpendicular to the bridge longitudinal axis.

The data collected by the SHM revealed that the VIVs of the bridge were dominated by vertical vibrations, in which the 1/8, 3/8, and 5/8 span of the two main spans vibrated more significantly than that of elsewhere. The preliminary analysis indicated that the time-frequency characteristics of the vertical vibrations of these locations were very similar, and thus the results of 3/8 span of Hanyang-side main span are presented for demonstration purpose. Fig. 7 shows the measured vertical displacement time history at 3/8 span of Hanyang-side main span on April 26, 2020. The two VIV events are clearly indicated in Fig. 7, and the maximum amplitude of the vertical displacement reached 0.50 m during the second VIV event. Fig. 8 further displays the measured vertical acceleration response at 3/8 span of Hanyang-side main span and the corresponding power spectral density function (PSD). As indicated in Fig. 8, the maximum acceleration was 0.75 m/s^2 and the 3/8 span of Hanyang-side main span had a single motional frequency of 0.244 Hz.

The Chinese code (JTG/T3360-01-2018, 2018) has specified the vibrational serviceability performance index during VIV, in which the vorticity limit of main girder is given by:

$$h_a = \frac{0.04}{f_b} \quad (12)$$

where h_a denotes the allowable amplitude of the VIV; f_b denotes the frequency of the VIV. The vibrational serviceability performance index is also adopted in Japanese guide for short (Sato, 2003).

In consideration that the VIV of a bridge generally presents simple harmonic vibration of a single mode, the displacement response of the vertical vortex vibration can be approximated by sine or cosine functions (Cao et al., 2020). In view of this, the allowable value of the corresponding acceleration amplitude can be calculated as:

$$A_a = \frac{0.04}{f_b} \omega_b^2 = \frac{0.04}{f_b} (4\pi^2 f_b^2) = 1.58 f_b \quad (13)$$

where A_a denotes the allowable acceleration amplitude of the VIV; and ω_b denotes the circular frequency of the VIV.

By substituting the frequency of the VIV of YWB, i.e., $f_b = 0.244 \text{ Hz}$, into Eqs. (12) and (13), the allowable amplitude and acceleration of VIV can be computed as $h_a = 0.164 \text{ m}$ and $A_a = 0.386 \text{ m/s}^2$. The observed displacement is three times larger than the allowable serviceability index, and the acceleration has doubled the allowable serviceability index.

Furthermore, the excited mode of YWB during VIV is identified using the mode shapes in finite-element analysis and video clips obtained from closed-circuit television (CCTV). Both the displacement along the bridge axis and the CCTV image indicated an asymmetrical vertical mode about

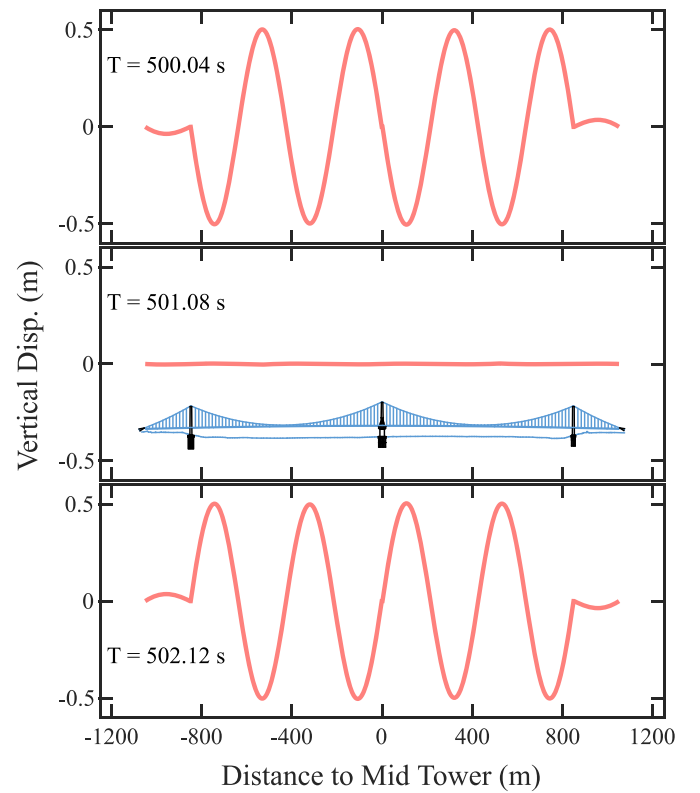


Fig. 10. Vertical bridge displacement along bridge under simulated VIV.

the intermediate tower. The three-dimensional (3D) FE model consisting of 1548 nodes and 2236 elements was established for the YWB using the FE software ANSYS, in which the bridge girder, tower, and pier were idealized as 3D beam elements, and the main cables and hangers were modelled with cable elements. Table 3 shows the first eight vertical modes along with mode shape descriptions of the YWB based on the FE analysis. In Table 3, VAS and VS are abbreviations for “vertical asymmetric” and “vertical symmetric”, respectively. The coincidence in measured motional frequency (0.244 Hz) and the calculated frequency for the fourth vertical asymmetric (VAS4) (0.242 Hz) mode demonstrates how the high potential of the VAS4 mode was triggered during the VIV event.

3.1.3. Identification of vortex-induced vibration

As mentioned above, large-amplitude VIVs were observed on YWB on April 26, 2020, which were locked in the fourth vertical asymmetric

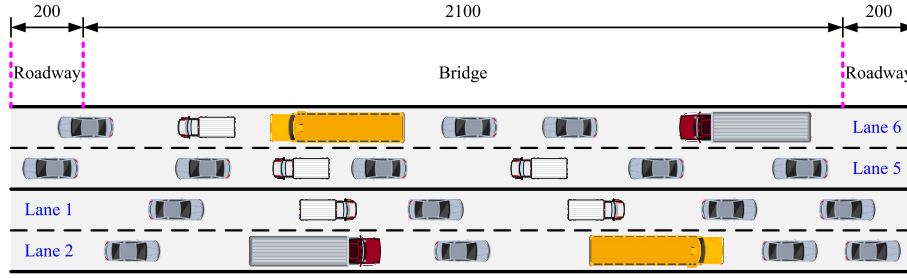
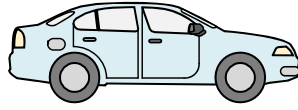
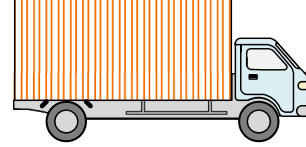


Fig. 11. Schematic of stochastic traffic flow in the roadway-bridge-roadway system (unit: m).

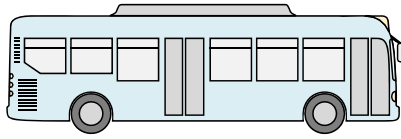
V1: Sedan Car



V2: Mini Van



V3: Motor Bus



V4: Truck

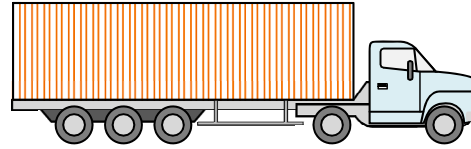


Fig. 12. Schematic of representative vehicles in the case study.

mode at a wind velocity of 7.0–7.6 m/s on the deck, and the maximum displacement and acceleration amplitudes are far beyond the allowable serviceability index specified in Chinese code (JTG/T3360-01-2018, 2018). Hence, in the present study, the observed VIV is adopted for the ride comfort evaluation of stochastic traffic flow crossing YWB experiencing VIV. According to the VIV modelling approach in subsection 2.1 and the field-monitoring data introduced above, the parameter of the VIF model is determined as follows.

Considering that the maximum amplitude of VIV was observed to be 0.50 m during VIV, the to-be-determined parameter $\tilde{C}_L$ in Eq. (1) can be conveniently determined through numerical simulations as follows.

$$\frac{a}{F_{Unit}} = \frac{\frac{1}{2} \rho U^2 D \tilde{C}_L}{F_{Unit}} = \frac{A_1}{A_0} \quad (14)$$

$$\mathbf{M}_b \ddot{\mathbf{u}}_b + \mathbf{C}_b \dot{\mathbf{u}}_b + \mathbf{K}_b \mathbf{u}_b = \mathbf{F}_{bw} \quad (15)$$

where $A_1 = 0.50$ m denotes the maximum amplitude of VIV; A_0 denotes the maximum amplitude of the deck under unit VIF F_{Unit} (1 kN in the present study), which is calculated through solving the governing equation of motion of the bridge (Eq. (15)) with the Newmark- β method. Based on the above procedure, the coefficient a is determined to be 51020.4 N. Subsequently, the determined VIF is applied to the bridge FEM for validation. Fig. 9 presents the simulated vertical displacement at 1/8 and 2/8 span of Hanyang-side main span under VIF over the time, and Fig. 10 shows the simulated vertical bridge displacement along the bridge at three special time steps. It is found that both the maximum amplitude (i.e., 0.50 m) and the frequency (i.e., 0.244 Hz) of the observed VIV event are well simulated through the proposed approach. Therefore, the stable VIV with a maximum amplitude of 0.50 m is further adopted for the ride comfort evaluation. Moreover, considering that the wind velocity maintained 7.0–7.6 m/s in the VIV event, a turbulent wind field composed of horizontal and vertical turbulence component is generated with a mean wind speed of 7.3 m/s to model the wind force on the vehicle, according to the spectra proposed by Panofsky and McCormick (1960) and Simiu and Scanlan (1996).

Table 4

Parameters of LAI-E model.

Parameter	Definition	Value			
N_{cell}	Number of cells occupied by the vehicle	V1	V2	V3	V4
v^{max}	Speed limit	10	16	16	28
		22 m/s	20 m/s		
v^s	Slow velocity	8 m/s	8 m/s		
a^{nor}	Normal acceleration/deceleration capability	2 m/s ²	1 m/s ²		
a^{emer}	Emergency deceleration capability	4 m/s ²	2 m/s ²		
R_0	Parameter of stochastic noise R_a	0.8	0.8		
R_d	Parameter of stochastic noise R_a	1	1		
R_s	Probability of random braking	0.01	0.01		
R_c	Probability of lane changing	1	1		

3.2. Prototype traffic

Since the dynamic response of the WTB system is significantly affected by the complex interactions between wind, stochastic traffic, and bridge, each component in the sophisticated system should be rationally modelled to derive a reliable conclusion with respect to the ride comfort evaluation. In view of this, the LAI-E model-based traffic simulation technique is performed to investigate the drivers' ride comfort on the YWB under VIV. Herein, a 2500 m-long roadway-bridge-roadway system (Fig. 11) is used to replicate the stochastic traffic flow on the bridge in a realistic way. Since the time duration of WTB coupled analysis is not too long (800 s in the present study), a periodic boundary, assuming the traffic density and the traffic proportion remain constant, is adopted (Chen and Wu, 2010, 2011; Xiong et al., 2020). Considering that the traffic control is usually performed during the VIV event of a long-span bridge, it is assumed that only half of the running capacity of YWB (i.e., Lane 1, 2, 5, 6 in Fig. 6) is open. Meanwhile, to reproduce real-world heterogeneous traffic, four types of vehicles (Fig. 12) are adopted in the present study. For brevity, the numerical model of the four types of vehicles and the vehicular dimensional and dynamic

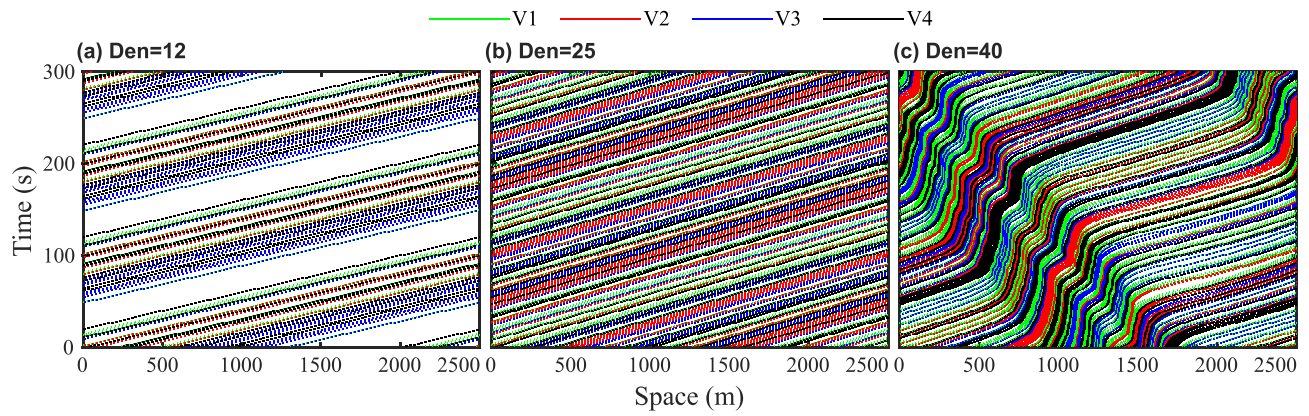


Fig. 13. Time-space maps of typical stochastic traffic flow.

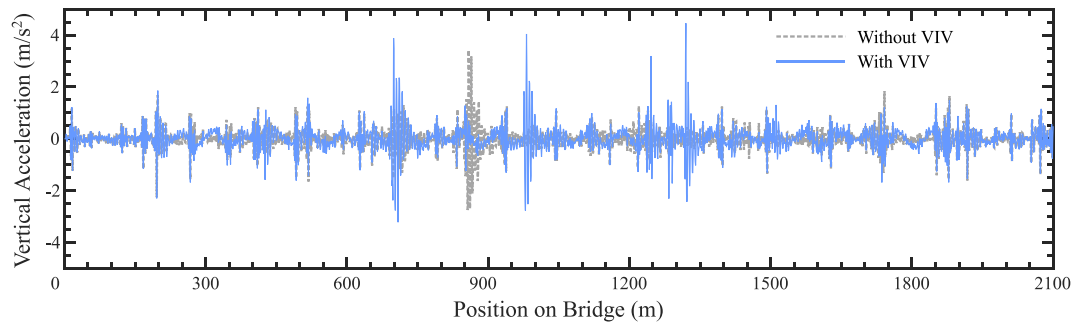


Fig. 14. Vertical acceleration response at seat location of typical sedan car with and without considering VIV of the bridge.

properties are listed in [Appendix A](#). The stochastic traffic flow is simulated through the LAI-E model based on the traffic density and proportion of interest according to the parameters listed in [Table 4](#) to meet empirical observations ([Li et al., 2016](#); [Guzmán et al., 2018](#)).

When simulating the traffic flow, the initial sequence and gap among all vehicles (i.e., initial state) are first randomly generated using the Monte Carlo method. Later, each vehicle in the initial state will be

updated at every time step based on rules in LAI-E model. As the traffic flow evolves, the running trajectories of typical stochastic flows under investigation are presented in [Fig. 13](#) in the form of time-space map, in which each line represents one vehicle at a different time in the roadway-bridge-roadway system. It can be seen from [Fig. 13\(a\)–\(c\)](#) that as the traffic density increases from 12 to 25, and 40 veh/km/lane, the obvious start-stop wave tends to emerge in the traffic flow, indicating

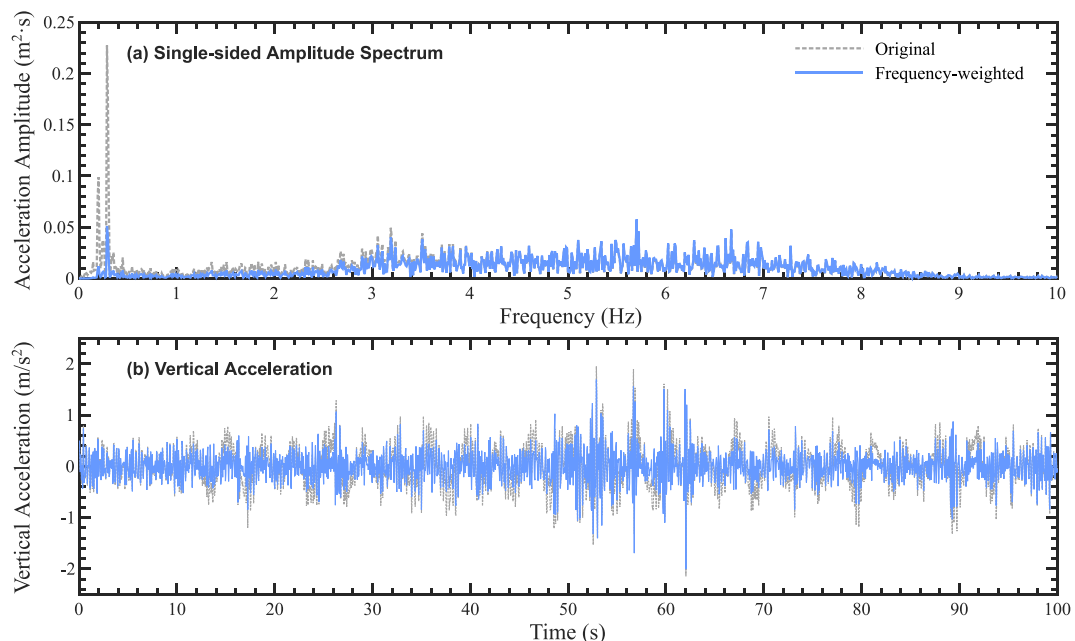


Fig. 15. Vertical acceleration response at seat location of typical sedan car: (a) single-side amplitude spectrum; (b) vertical acceleration.

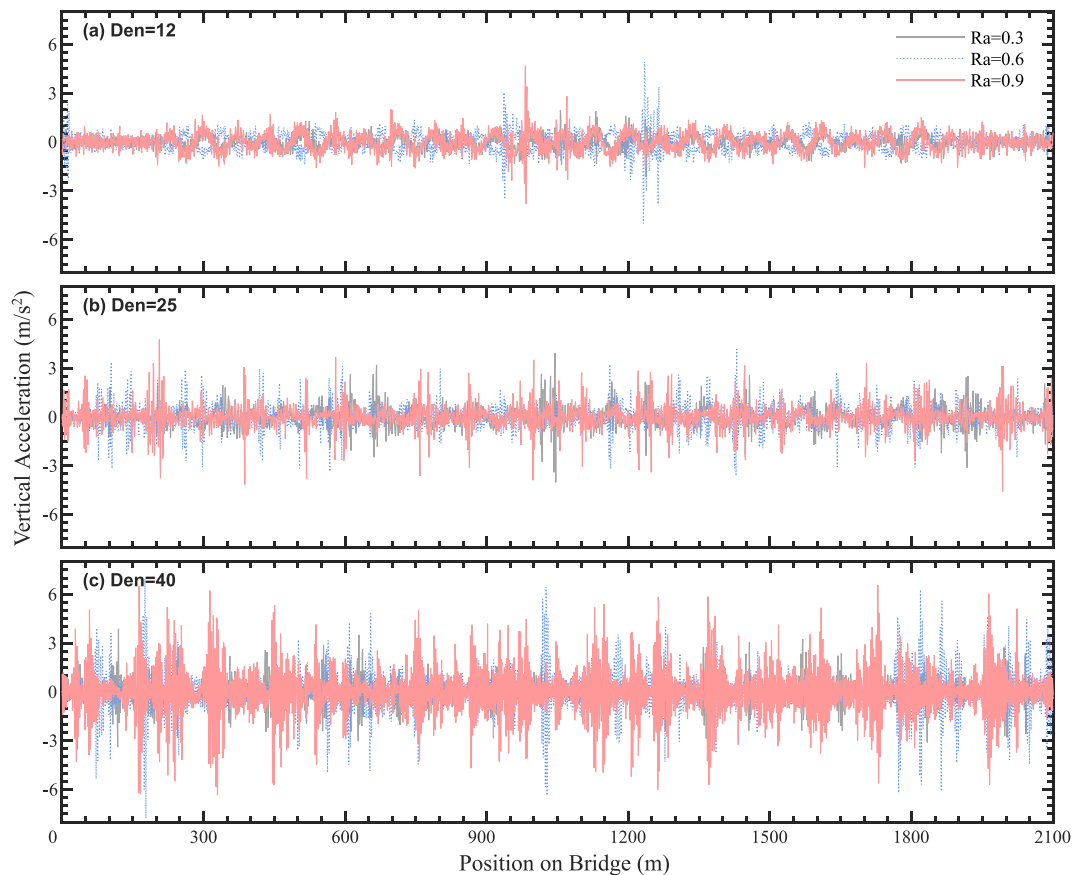


Fig. 16. Vertical acceleration at seat location of typical sedan car under different scenarios.

the transmission of traffic congestion in the opposite direction. As a result, these three traffic densities are adopted to study the ride comfort under different traffic conditions (free, moderate, and busy traffic conditions) subsequently.

4. Analytical result

As illustrated previously, the VIV event with the maximum amplitude of 0.50 m and the frequency of 0.244 Hz is adopted to evaluate the ride comfort of stochastic traffic flow crossing YWB experiencing VIV. Meanwhile, the traffic flow is generated based on predefined traffic density and proportion. In the present study, three types of density (Den = 12, 25, and 40 veh/km/lane) and three types of proportion (Ra = 0.3, 0.6, and 0.9) are considered. Note that the traffic proportion (i.e., Ra) means the ratio of the heavy vehicle (V2, V3, and V4) amount to the total vehicle amount, and the proportion of each type of heavy vehicle is the same under any Ra. For instance, when Ra = 0.3, the proportion of V1, V2, V3, and V4 in each lane is 0.7, 0.1, 0.1, and 0.1, respectively. Furthermore, the road roughness is generated through an average of 20-time simulations according to the road roughness condition in Eq. (3) (1280×10^{-6} , 320×10^{-6} , 80×10^{-6} , 20×10^{-6} , and 5×10^{-6} m³/cycle for the A, B, C, D, and E level) given its uncertain nature. The turbulent wind field ($U = 7.3$ m/s) for vehicle wind load analysis keeps the same among each analysis. To get a stable and reliable result of the WTB system, the total simulation duration is 800 s with a time step of 0.04 s (Chen and Wu, 2010, 2011). It should be noted that although an 800 s-long simulation allows an adequate evolution of the traffic flow to reflect various traffic distributions on the bridge, the different initial states of traffic flow may also have influences on the result. However, lots of computational efforts are required to investigate such an influence, which is beyond the scope of this study and is worth further

Table 5

RMS value of typical sedan car crossing YWB under VIV in different scenarios (m/s²).

Den (veh/km/lane)	12	25	40
Ra			
0.3	0.39	0.59	0.68
0.6	0.57	0.71	1.13
0.9	0.59	0.75	1.36

endeavors.

4.1. Dynamic response of vehicle

It is found in the preliminary research of the authors (Zhu et al., 2018a) that the vertical acceleration at the seat location plays the most significant role in drivers' ride comfort compared to the responses at other locations and directions. Hence, in this subsection, the vehicle vertical acceleration at the seat is adopted to illustrate the ride comfort evaluation process as well as the vehicle dynamic characteristics under different scenarios.

To highlight the effect of VIV on drivers' rider comfort, a comparison of vehicle responses with and without considering VIV is first conducted. For better comparison, the wind effect is assumed to act on the vehicle but not on the bridge when the VIV is not considered. Meanwhile, the moderate traffic flow (Den = 25 veh/km/lane) with Ra = 0.9 and A-level road roughness condition are adopted. As shown in Fig. 14, the maximum absolute value and fluctuation of vertical acceleration response of a typical sedan car get larger after considering VIV, which can make the driver feel less comfortable. In specific, for the selected sedan car, the RMS of vertical acceleration increases 32% from 0.38 m/

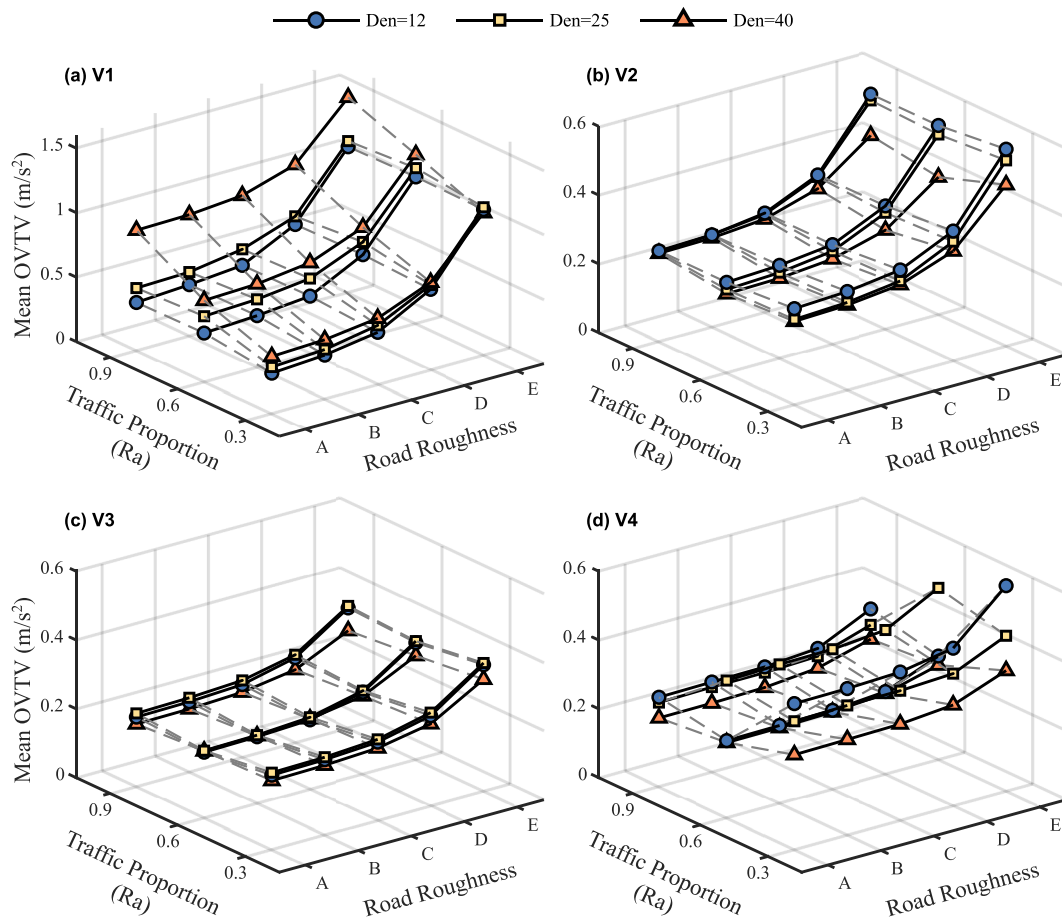


Fig. 17. Mean value of OVTV among different scenarios.

s^2 to 0.50 m/s^2 after considering VIV.

As mentioned above, the vehicle acceleration response should be frequency-weighted before ride comfort evaluation. Herein, a typical sedan car in the traffic flow (Den = 12 veh/km/lane, and Ra = 0.3) is taken as an illustrative example. Fig. 15 presents the vertical acceleration at seat location before and after frequency-weighting of a typical sedan car crossing YWB under VIV. It is obvious that the vertical acceleration response reduces in both frequency and time domain. The maximum acceleration amplitude reduces 74.61% from 0.23 to 0.06 m/s^2 , and the maximum absolute acceleration reduces 6.47% from 2.15 to 2.01 m/s^2 . Moreover, the RMS of vertical acceleration has a reduction of 23.58% from 0.39 to 0.30 m/s^2 . Since the frequency-weighted response may significantly differ from the original response, the frequency-weighting procedure is of vital importance in conducting a rational ride comfort evaluation.

To better understand the vibration characteristics of the vehicle and to evaluate the ride comfort in a comprehensive perspective, the vertical acceleration of a typical sedan car crossing YWB experiencing VIV under different scenarios is presented in Fig. 16, and the corresponding RMS values are summarized in Table 5. It can be seen from the time histories and RMS values that the fluctuation of vehicle vertical acceleration crossing YWB tends to increase as the traffic density and the proportion of heavy vehicles in the traffic flow go higher. For instance, when Ra = 0.3, the RMS value of the representative sedan car in the traffic flow with a density of 12, 25, and 40 veh/km/lane is 0.39, 0.57, and 0.59, respectively. As the Ra increases from 0.3 to 0.6 and 0.9, the selected sedan car in free traffic flow (Den = 12 veh/km/lane) has the RMS value of 0.39, 0.59, and 0.68, respectively. This may be due to that the larger vehicle load triggers more excessive bridge dynamic response through complex interactions, and therefore the vehicle runs bumpily when

crossing the bridge.

4.2. Drivers' ride comfort evaluation

In this subsection, the ride comfort of stochastic traffic flow crossing YWB experiencing VIV is assessed by the OVTV of the vehicles based on the ISO 2631-1 recommendation (ISO, 1997). Specifically, every time each vehicle in the traffic flow crosses the bridge, the time-history vehicle acceleration response is collected, and the corresponding OVTV is calculated. Under the assumption of periodic boundary, each vehicle may cross the bridge many times during the entire simulation time (800 s), generating several OVTV values for ride comfort evaluation. In particular, the influence of traffic density, traffic proportion, and road roughness condition on the drivers' ride comfort will be addressed through 45 different WTB simulations (3 kinds of traffic density $\times$ 3 kinds of traffic proportion $\times$ 5 kinds of road roughness).

4.2.1. Mean value of OVTV

As mentioned above, the OVTV values of all vehicles in the traffic flow are obtained, where several journeys that each vehicle crosses the bridge are treated as independent samples. In this way, thousands of OVTV samples can be obtained through a 800 s-long simulation. For different types of vehicles under investigation, the vibration characteristics and associated ride comfort vary a lot. Therefore, the mean value of each type of vehicle's OVTV samples in a simulation is calculated and presented in Fig. 17. It is seen from Fig. 17(a) that the mean OVTV of sedan cars increases dramatically when the traffic density (Den) and proportion of heavy vehicles (Ra) go higher, which coincides with the observation on RMS values mentioned above. The increase of mean OVTV caused by larger Ra is more significant when the traffic density is

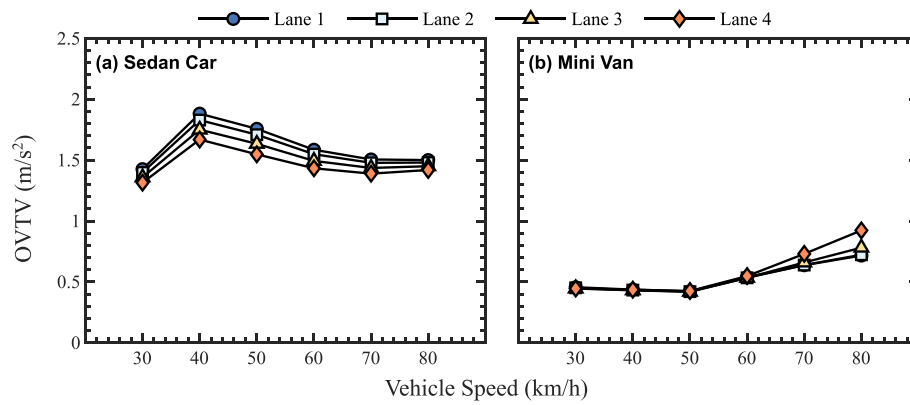


Fig. 18. OVTV of single sedan car and single mini van crossing the bridge under VIV.

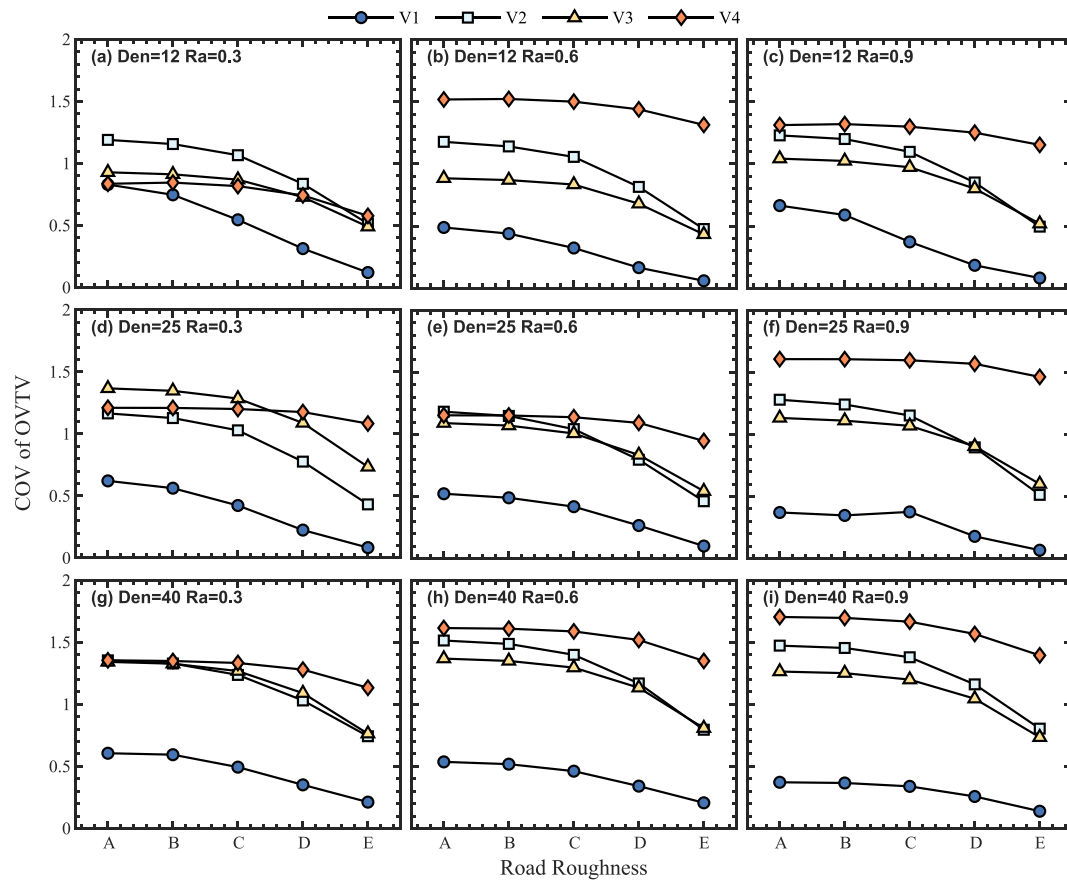


Fig. 19. Coefficient of variance of OVTV among different scenarios.

large, which may be because a more violent bridge vibration is induced by the heavier vehicle load. Meanwhile, it is straightforward that as the road roughness condition gets poorer (from Level A to E), the mean value of OVTV gets higher. In particular, when the road roughness condition varies from Level A to C, the increase of mean OVTV is moderate. However, as the road roughness condition gets worse and beyond the average (Level C), the mean OVTV increases obviously. For instance, for the sedan car in the traffic flow with a density of 12 veh/km/lan and a proportion of 0.3, the mean OVTV is 0.31, 0.34, 0.40, 0.62, and 1.13 m/s^2 separately under the road roughness condition from

Level A to E. This suggests that the importance of regular road maintenance to ensure the ride comfort of vehicles when crossing the bridge.

Similar patterns can also be found in the mean OVTV of three types of heavy vehicles (V2, V3, and V4), where the road roughness can affect the mean OVTV directly. Nevertheless, the traffic density and proportion have a relatively inapparent effect on the mean OVTV of heavy vehicles, and the mean OVTV of heavy vehicles is smaller than the sedan car. In specific, for the V2 and V3, the mean OVTV is seldom influenced by traffic density and proportion. Take the mini van in the traffic flow under average road roughness condition (Level C) as an example. When

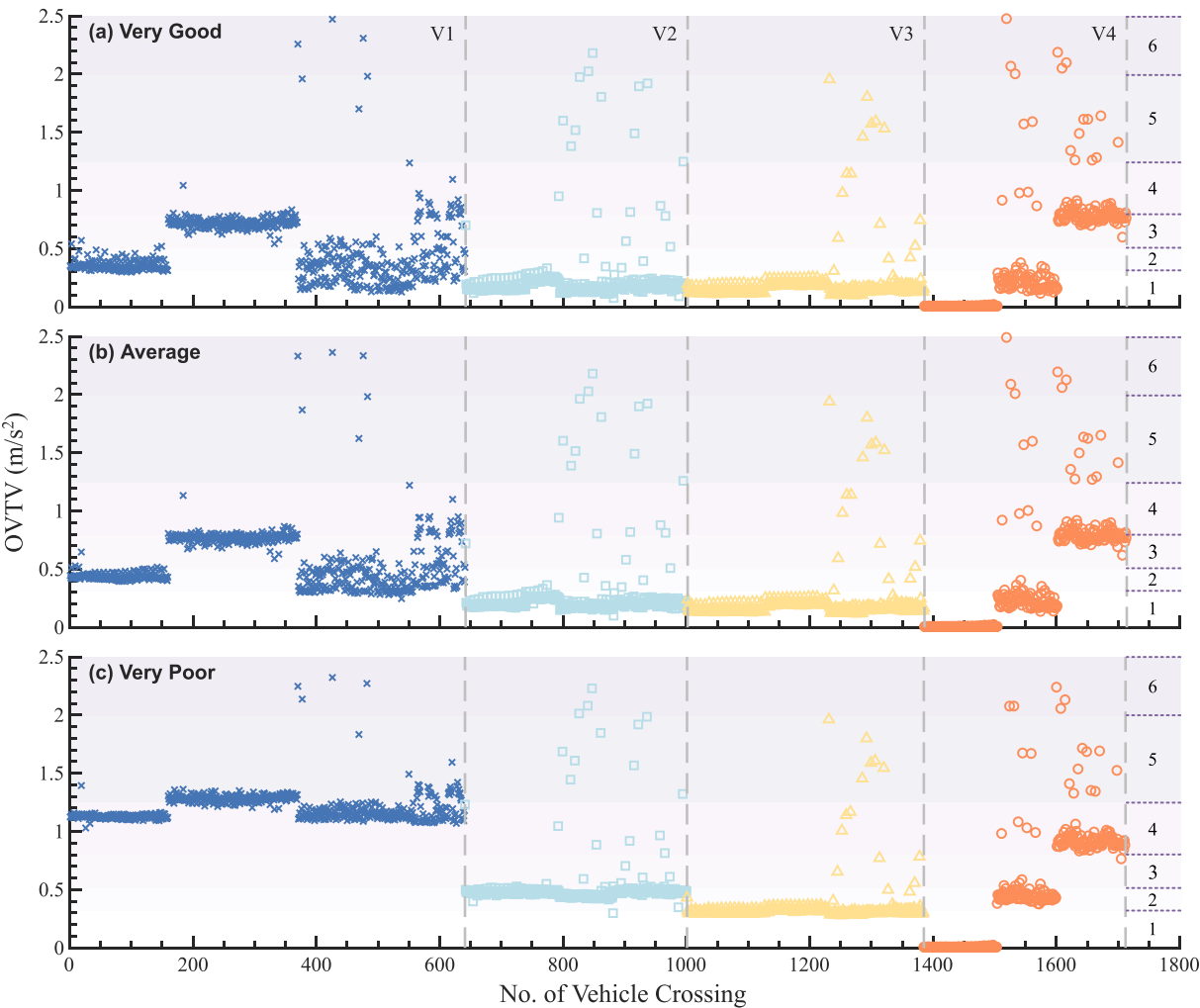


Fig. 20. OVTV of all vehicles in one traffic flow crossing bridge under different road roughness conditions.

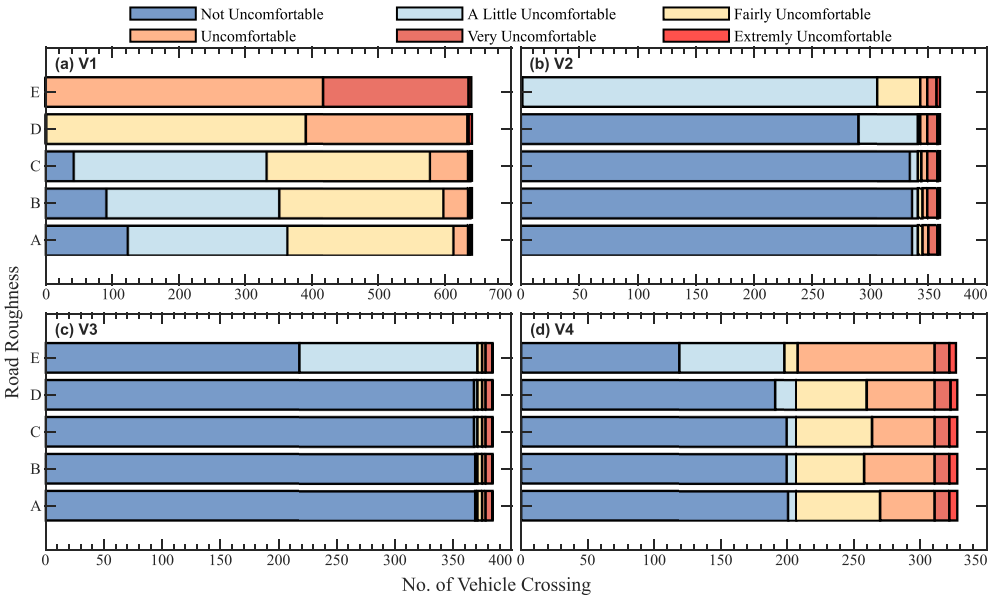


Fig. 21. Comfort level of all types of vehicles in a traffic flow crossing bridge under different road roughness conditions.

Table 6
Comfort level of sedan car crossing bridge (%).

Traffic Proportion (ra)		0.3					0.6					0.9				
Roughness		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
Comfort Level																
Den = 12	1	71.8	65.7	16.5	0.0	0.0	43.6	39.4	11.4	0.0	0.0	65.3	61.3	24.0	0.0	0.0
	2	18.4	24.3	76.2	0.0	0.0	32.9	33.9	63.8	0.0	0.0	4.0	6.7	52.0	0.0	0.0
	3	5.0	4.8	2.4	94.8	0.0	21.8	25.1	22.8	90.6	0.0	22.7	24.0	21.3	89.3	1.3
	4	3.3	3.6	3.3	3.6	94.8	1.3	1.3	1.6	9.1	92.5	8.0	8.0	2.7	10.7	89.3
	5	1.0	0.9	1.0	0.7	4.3	0.3	0.3	0.3	0.3	7.5	0.0	0.0	0.0	0.0	9.3
	6	0.5	0.7	0.5	0.9	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Den = 25	1	60.4	57.3	24.2	0.2	0.0	19.3	14.4	6.7	0.0	0.0	14.8	11.5	2.5	0.0	0.0
	2	6.3	8.0	40.0	0.1	0.0	37.3	40.4	45.1	0.3	0.0	45.9	54.1	51.6	0.0	0.0
	3	30.9	32.1	33.0	92.6	0.1	38.8	38.4	38.2	60.7	0.0	33.6	29.5	33.6	77.9	0.0
	4	1.7	1.9	2.1	6.4	95.4	3.6	5.9	9.0	37.9	65.2	5.7	4.9	12.3	22.1	81.1
	5	0.4	0.4	0.4	0.6	4.3	0.5	0.5	0.5	0.3	34.2	0.0	0.0	0.0	0.0	18.9
	6	0.2	0.2	0.2	0.2	0.2	0.5	0.5	0.5	0.8	0.6	0.0	0.0	0.0	0.0	0.0
Den = 40	1	44.3	34.8	14.9	1.9	0.3	13.7	13.1	6.5	0.6	0.0	0.0	0.0	0.0	0.0	0.7
	2	22.9	31.9	48.2	15.1	0.6	43.5	38.8	15.3	10.0	0.3	2.0	2.0	1.3	0.0	0.0
	3	26.8	27.1	30.4	54.2	5.7	12.3	17.8	47.3	49.9	2.1	54.6	53.3	46.7	3.9	0.0
	4	4.7	4.6	5.0	27.2	66.0	26.8	25.5	24.6	25.7	35.8	21.7	21.1	21.7	60.5	0.7
	5	1.0	1.2	1.1	1.1	26.5	3.4	4.7	6.0	13.4	61.2	20.4	22.4	28.9	34.2	97.4
	6	0.4	0.5	0.4	0.5	0.9	0.3	0.2	0.3	0.3	0.6	1.3	1.3	1.3	1.3	1.3

the traffic density is 25 veh/km/lane, the mean OVTV of mini van in the traffic flow with $Ra = 0.3, 0.6$, and 0.9 is $0.27, 0.27$, and 0.28 m/s^2 . While the traffic proportion is 0.3 , the mean OVTV of V2 in the 12, 25, and 40 veh/km/lane traffic flow is $0.30, 0.27$, and 0.26 m/s^2 . These may be owing to two reasons. Firstly, the dynamic property of the selected sedan car (i.e., the model parameters) makes its driver more likely to feel uncomfortable than the driver of heavy vehicles. To show this, the comparison of the OVTV of a single sedan car and a single mini van crossing the bridge under VIV is conducted. In the comparison, the single vehicle (sedan car or mini van) is assumed to cross the bridge under VIV through different lanes (Lane 1–4 in Fig. 6) with a constant speed (6 kinds of speed are considered for better illustration). The OVTV of the single vehicle crossing bridge through each lane is calculated and shown in Fig. 18. It is found that the OVTV of sedan car is larger than that of mini van crossing the bridge under VIV, even with the same running trajectory and speed. Similar results can be also found in Ref. (Zhou and Chen, 2016), where the OVTV of sedan car crossing a bridge is larger than that of heavy truck. Then, the larger speed limit and acceleration/deceleration ability of sedan car may also have a connection to these results. It is noted that the mean OVTV of trucks in Fig. 17 (d) shows a distinct feature with regards to traffic density and proportion, which will be illustrated later.

4.2.2. Coefficient of variance of OVTV

To further understand the ride comfort of traffic flow crossing the bridge experiencing VIV, the coefficient of variance (COV) of all OVTV samples for a specific type of vehicle in a simulation is calculated and presented in Fig. 19. It is found that the COV decreases as the road roughness condition getting worse. Meanwhile, the variation of COV with traffic density and proportion is insignificant, especially for V1, V2, and V3. For example, under the Level C roughness condition, the COV of V1 in 25 veh/km/lane traffic flow is $0.42, 0.41$, and 0.38 for $Ra = 0.3, 0.6$, and 0.9 ; when $Ra = 0.3$, the COV of V1 is $0.55, 0.42$, and 0.49 in the traffic flow with density of 12, 25, and 40 veh/km/lane. These suggest that when the road roughness condition gets worse, the drivers' ride comfort samples under a same traffic flow tends to maintain a stable status, and the randomness of drivers' ride comfort samples among different simulations (with the same road roughness but different traffic flows) barely varies.

4.2.3. Comfort level

In this subsection, the comfort level of each type of vehicle in the traffic flow crossing YWB under VIV is assessed according to Table 2. In detail, all the OVTV samples obtained in the simulations are categorized to different comfort levels, in which the lower limit of OVTV range in Table 2 is treated as the threshold. Fig. 20 illustrates all the OVTV samples in a typical traffic flow (Den = 25 veh/km/lane, $Ra = 0.6$) under different road roughness conditions. It is also found that the randomness of OVTV samples gets lower as the road roughness gets worse. Particularly, three obvious clusters can be seen in the OVTV samples of V4, presenting a unique feature as shown in Fig. 17. In fact, the three clusters are caused by the different vehicle vibration characteristics in various lanes. The first cluster near zero is referred to the vehicles travelling on Lane 1 and 2 in Fig. 6, and the second and third clusters are related to the vehicles on Lane 5 and 6, respectively. The similar phenomenon is also found in previous research of the author (Zhu et al., 2018a), indicating the different comfort levels of vehicles running on different lanes of the bridge. These may be caused by the different bridge responses at each lane and the various wind effect of vehicle on different lanes. Notably, the front area of V4 is much larger than the other three types of vehicles, and this will lead to a distinct wind effect. However, it remains challenging to derive a comprehensive view on the unique feature, which needs further endeavors in the future.

Furthermore, the sample amounts of vehicles belong to different comfort levels in each simulation are summarized. For instance, the comfort levels of four types of vehicles in a traffic flow (Den = 25 veh/km/lane, $Ra = 0.6$) crossing the bridge under different road roughness conditions are presented in Fig. 21. It is obvious that as the road roughness goes worse, the drivers feel more uncomfortable, especially when the road roughness condition is beyond the average (Level C). Meanwhile, the driver of a sedan car is more likely to feel uncomfortable compared to the driver of the other three types of vehicles. These observations are also similar to those from the mean OVTV. Similar patterns can be found in the comfort levels of each type of vehicle under different scenarios in percentage as well. In specific, as listed in Table 6, when the road roughness gets worse from Level A to E, the discomfort probability (i.e., the proportion of vehicles in the status of level 2, 3, 4, 5, and 6) sedan cars in the traffic flow (Den = 12 veh/km/lane, $Ra = 0.3$) is $28.2\%, 34.3\%, 83.5\%, 100\%$, and 100% , respectively. Moreover, the discomfort probability of sedan car also increases as the traffic density

and proportion increase. As limited by space, only the comfort levels of sedan cars are shown in Table 6. The comfort levels of the other three types of vehicles are referred to Appendix B.

5. Concluding remarks

This paper presents a general ride comfort evaluation procedure for the stochastic traffic flow crossing a long-span bridge under vortex-induced vibration (VIV) for the first time, which is achieved by a framework of wind-traffic-bridge (WTB) coupled analysis developed by the authors. The VIV of the bridge is simulated by a harmonic VIF model based on field-monitoring data. Several typical traffic flows are generated through a microscopic cellular automaton traffic model, and the ride comfort of all vehicles in the traffic flow crossing a prototype long-span suspension bridge experiencing VIV is assessed. The following conclusions are drawn from the present study.

- (1) With the proposed WTB analytical framework and harmonic VIF model, the dynamic response of the traffic flow running on a long-span bridge experiencing VIV can be properly simulated, which provides a fundamental basis for the ride comfort evaluation. The developed procedure can be easily applied to other long-span bridges suffered from the VIV problem.
- (2) The frequency-weighting procedure is crucial in conducting ride comfort evaluation. For a typical sedan car, the maximum acceleration amplitude reduces 74.61% from 0.23 to 0.06 m/s^2 , and the maximum absolute acceleration reduces 6.47% from 2.15 to 2.01 m/s^2 after frequency weighting.
- (3) The ride comfort of sedan cars in the traffic flow crossing the bridge under VIV obviously increases with the traffic density and proportion of heavy vehicles. Meanwhile, as the road roughness condition gets poorer, the driver of a sedan car tends to feel more uncomfortable. For instance, for the sedan car in the traffic flow with a density of 12 veh/km/lane and a heavy vehicle proportion of 0.3, the mean overall vibration total value (OVTV) is 0.31, 0.34, 0.40, 0.62, and 1.13 m/s^2 separately under the road roughness condition from Level A to E.
- (4) The ride comfort of mini van, motor bus, and truck also gets lower with the poor road roughness, but it is seldom affected by the traffic density and proportion of heavy vehicles (R_a). Take the

mini van in the traffic flow under average road roughness condition (Level C) as an example. When the traffic density is 25 veh/km/lane, the mean OVTV of mini van in the traffic flow with $R_a = 0.3, 0.6$, and 0.9 is 0.27, 0.27, and 0.28 m/s^2 . While $R_a = 0.3$, the mean OVTV of mini van in the 12, 25, and 40 veh/km/lane traffic flow is 0.30, 0.27, and 0.26 m/s^2 .

- (5) For all the drivers' ride comfort samples obtained from each time a vehicle crossing the bridge in a simulation, the samples tend to maintain a stable status as the road roughness condition gets worse, and the randomness of drivers' ride comfort samples among different simulations (with the same road roughness but different traffic flows) barely varies. For example, under the Level C roughness condition, the coefficient of variance (COV) of the OVTV of all sedan cars in 25 veh/km/lane traffic flow is 0.42, 0.41, and 0.38 for $R_a = 0.3, 0.6$, and 0.9 ; when $R_a = 0.3$, the COV of the OVTV of all sedan cars is 0.55, 0.42, and 0.49 in the traffic flow with density of 12, 25, and 40 veh/km/lane.

Credit authors contribution statement

Jin Zhu: Conceptualization, Methodology, Writing - original draft & review & editing, Supervision. **Ziluo Xiong:** Methodology, Software, Formal analysis, Visualization, Writing - original draft. **Huoyue Xiang:** Writing - review & editing. **Xu Huang:** Validation, Data curation. **Yongle Li:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Numerical model and dynamic/dimensional parameter of representative vehicle

Table A.1

Definition of vehicle numerical model parameter

Parameter	Definition	Parameter	Definition
I_{yvi}	Pitching moment of inertia of rigid body	M_{vri}	Mass of the rigid body
I_{xvi}	Yawing moment of inertia of rigid body	M_{valj}	Mass of axle block
*L_a	Dimensional parameter	I_{xvi}	Rolling moment of inertia of rigid body
K_{uzj}	Upper vertical spring stiffness	C_{uzj}	Upper vertical damping coefficient
K_{lyj}	Upper lateral spring stiffness	C_{lyj}	Upper lateral damping coefficient
K_{lsj}	Lower vertical spring stiffness	C_{lsj}	Lower vertical damping coefficient
K_{lyj}	Lower lateral spring stiffness	C_{lyj}	Lower lateral damping coefficient
h_{vi}	Reference height	A_f	Frontal area

Note: (1) for two-axle vehicle (V1, V2, and V3): $i = 1, j = 4$; for five-axle vehicle (V4): $i = 2, j = 10$; (2) L_a can be found in the schematic of vehicle numerical model.

Two-axle vehicle

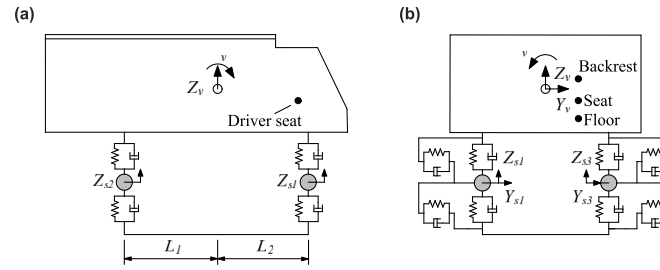


Fig. A.1. Numerical model of two-axle vehicle: (a) Elevation view. (b) Side view.

Table A.2

Main properties and dimensions of two-axle vehicle

Parameter	Unit	Sedan Car	Mini Van	Motor Bus
M_{yr}	kg	1280	6000	12190
I_{yv}	kg·m ²	2000	34000	79000
I_{xv}	kg·m ²	560	2950	13000
I_{zv}	kg·m ²	32000	12500	53000
M_{yvalj}	Kg	200/160	800/700	340/525
K_{usz}	10 ³ N/m	310/310	200/225	126/255
C_{usz}	N·s/m	6229/2000	3850/2720	9150/18300
K_{uyj}	10 ³ N/m	110/110	300/300	50/85
C_{uyj}	N·s/m	6229/2000	3850/2720	6000/6000
K_{luj}	10 ³ N/m	310/310	363/422	750/1500
C_{luj}	N·s/m	600/600	3000/3000	2000/2000
K_{lyj}	10 ³ N/m	110/110	120/120	300/600
C_{lyj}	N·s/m	600/600	3000/3000	2000/2000
L_1	m	1.7	2.6	3.4
L_2	m	1.4	3.0	1.8
h_{v1}	m	0.72	1.50	1.35
A_f	m ²	2.5	8.0	20.0

Five-axle vehicle

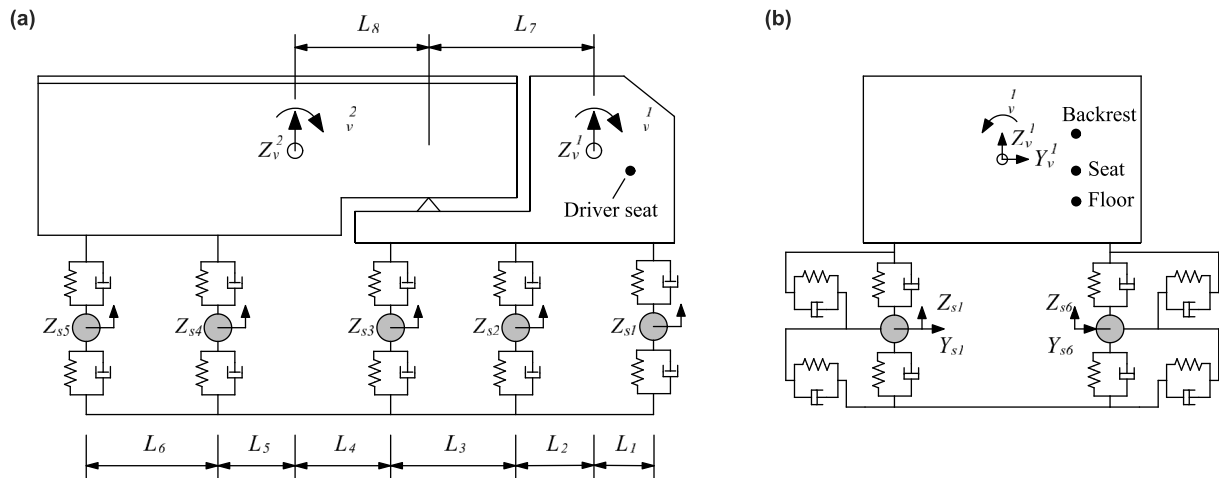


Fig. A.2. Numerical model of five-axle vehicle: (a) Elevation view. (b) Side view.

Table A.3
Main properties and dimensions of five-axle vehicle

Parameter	Unit	Truck
M_{vri}	kg	2276.5/45245
I_{yvi}	kg·m ²	20196/285990
I_{xvi}	kg·m ²	2189.2/43512
I_{zvi}	kg·m ²	287000/295000
M_{valj}	Kg	700/1000/1000/800/800
K_{uzj}	10 ³ N/m	300/1000/1000/1250/1250
C_{uzj}	10 ² N·s/m	100/530/530/530/530
K_{iyj}	10 ³ N/m	299/299/299/299/299
C_{iyj}	10 ² N·s/m	50/50/50/50/50
K_{tj}	10 ³ N/m	1500/3000/3000/3000/3000
C_{tj}	10 ² N·s/m	30/30/30/30/30
K_{lyj}	10 ³ N/m	260/260/260/260/260
C_{lyj}	10 ² N·s/m	20/20/20/20/20
L_1	m	1.0
L_2	m	2.0
L_3	m	1.4
L_4	m	3.8
L_5	m	3.2
L_6	m	1.4
L_7	m	2.7
L_8	m	4.5
h_{v1}	m	1.2
h_{v2}	m	1.7
A_f	m ²	25.0

Appendix B. Comfort level for different vehicles

Table B.1

Comfort level of mini van crossing bridge (%)

Traffic Proportion (ra)		0.3					0.6					0.9				
Roughness		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
Comfort Level																
Den = 12	1	88.2	87.4	86.6	77.3	0.0	95.4	95.4	94.9	76.4	0.0	88.9	91.6	87.5	70.4	0.0
	2	6.7	6.7	5.9	15.1	63.0	0.0	0.0	0.4	19.0	56.1	7.3	4.5	8.7	25.8	53.3
	3	0.8	1.7	3.4	3.4	30.3	0.8	0.8	0.8	0.8	40.1	0.0	0.0	0.0	0.0	42.9
	4	0.0	0.0	0.0	0.0	2.5	0.0	0.0	0.0	0.0	0.0	0.3	0.3	0.3	0.3	0.3
	5	4.2	4.2	4.2	4.2	2.5	3.0	3.0	3.0	2.5	2.5	2.4	2.4	2.4	2.1	2.1
	6	0.0	0.0	0.0	0.0	1.7	0.8	0.8	0.8	1.3	1.3	1.0	1.0	1.0	1.4	1.4
Den = 25	1	93.6	93.6	93.6	84.1	0.0	93.3	93.3	92.8	80.6	0.3	93.1	92.7	90.4	71.0	0.0
	2	1.3	1.3	1.3	10.8	67.5	1.4	1.4	1.9	14.2	84.7	1.9	2.3	4.2	23.7	77.5
	3	0.6	0.6	0.6	0.6	28.0	1.1	1.1	0.8	0.6	10.3	1.1	1.1	1.4	1.4	18.4
	4	1.3	1.3	1.3	1.3	1.3	1.4	1.1	1.4	1.7	1.7	0.5	0.5	0.5	0.5	0.9
	5	3.2	3.2	3.2	3.2	3.2	2.2	2.5	2.5	2.5	2.2	2.1	2.1	2.1	2.1	2.0
	6	0.0	0.0	0.0	0.0	0.0	0.6	0.6	0.6	0.6	0.8	1.2	1.2	1.2	1.2	1.2
Den = 40	1	87.5	87.5	87.5	86.9	39.2	90.3	90.3	90.3	90.0	49.9	88.9	88.9	88.9	88.5	47.5
	2	1.1	1.1	1.1	1.1	48.9	1.1	1.1	1.1	1.4	41.2	0.9	1.3	0.9	1.3	41.9
	3	7.4	7.4	7.4	3.4	1.7	1.7	1.7	1.4	1.4	1.1	1.8	1.6	1.8	1.3	1.8
	4	1.1	0.6	0.6	5.1	2.3	2.8	2.8	3.1	2.8	3.1	3.8	3.3	2.9	2.7	2.4
	5	2.3	2.8	2.8	2.8	7.4	3.9	3.9	3.9	4.2	4.2	3.8	4.0	4.7	5.3	5.3
	6	0.6	0.6	0.6	0.6	0.6	0.3	0.3	0.3	0.3	0.6	0.9	0.9	0.9	0.9	1.1

Table B.2

Comfort level of motor bus crossing bridge (%)

Traffic Proportion (ra)		0.3					0.6					0.9				
Roughness		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
Comfort Level																
Den = 12	1	96.1	96.1	95.1	92.2	29.4	97.5	97.5	97.5	97.5	49.4	94.8	94.8	94.8	94.8	63.9
	2	0.0	0.0	1.0	3.9	62.7	0.0	0.0	0.0	0.0	48.1	0.4	0.4	0.4	0.4	31.3
	3	0.0	0.0	0.0	0.0	3.9	0.6	0.6	0.6	0.6	0.0	2.0	2.0	2.0	2.0	1.6
	4	2.9	2.9	2.9	2.9	2.9	0.6	0.6	0.6	0.6	1.2	1.2	1.2	1.2	1.2	1.6
	5	1.0	1.0	1.0	1.0	1.0	1.2	1.2	1.2	1.2	1.2	1.6	1.6	1.6	1.6	1.6
	6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Den = 25	1	94.3	94.3	93.6	93.6	63.7	96.1	96.1	95.8	95.8	56.8	94.2	94.2	94.0	94.0	60.9
	2	0.0	0.0	0.6	0.6	29.9	0.5	0.5	0.8	0.8	39.8	0.4	0.4	0.6	0.6	33.7

(continued on next page)

Table B.2 (continued)

Traffic Proportion (ra)		0.3					0.6					0.9				
Roughness		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
Comfort Level																
Den = 40	3	0.6	0.6	0.6	0.6	1.3	1.0	1.0	1.0	1.0	1.0	1.7	1.7	1.7	1.7	1.4
	4	2.5	2.5	2.5	2.5	2.5	0.8	0.8	0.8	0.8	0.8	2.3	2.3	2.3	2.3	2.5
	5	1.9	1.9	1.9	1.9	1.9	1.6	1.6	1.6	1.6	1.6	0.8	0.8	0.8	0.8	1.0
	6	0.6	0.6	0.6	0.6	0.6	0.0	0.0	0.0	0.0	0.0	0.6	0.6	0.6	0.6	0.6
	1	94.7	94.7	94.7	94.7	72.4	92.9	92.9	92.9	92.9	92.1	93.8	93.8	93.6	93.6	91.2
	2	0.7	0.7	0.7	0.7	23.0	1.5	1.5	1.5	1.5	1.8	1.3	1.3	1.5	1.5	3.7
	3	0.7	0.7	0.7	0.7	0.0	1.3	1.0	1.0	1.0	1.5	0.6	0.6	0.6	0.6	0.7
	4	2.6	2.6	2.6	2.6	3.3	1.5	1.8	1.8	1.8	1.8	3.2	3.2	3.4	3.2	3.2
	5	0.7	0.7	0.7	0.7	0.7	2.8	2.8	2.8	2.8	2.8	0.9	0.9	0.7	0.9	0.9
	6	0.7	0.7	0.7	0.7	0.7	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.2	0.2

Table B.3

Comfort level of truck crossing bridge (%)

Traffic Proportion (ra)		0.3					0.6					0.9				
Roughness		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
Comfort Level																
Den = 12	1	64.9	64.9	64.9	54.4	10.5	79.0	79.0	79.0	76.5	58.8	73.2	74.1	73.2	69.6	53.1
	2	7.0	7.0	7.0	17.5	47.4	0.0	0.0	0.0	2.5	20.2	0.9	0.0	0.9	4.5	15.6
	3	8.8	3.5	8.8	7.0	14.0	13.4	11.8	14.3	12.6	2.5	18.3	15.2	17.9	16.1	5.8
	4	17.5	22.8	17.5	19.3	26.3	5.9	7.6	5.0	6.7	16.8	7.1	10.3	7.6	9.4	25.0
	5	1.8	1.8	1.8	1.8	1.8	1.7	1.7	1.7	1.7	1.7	0.4	0.4	0.4	0.4	0.4
	6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Den = 25	1	61.3	61.3	61.3	58.6	47.6	61.3	61.0	61.0	58.2	36.4	73.6	73.8	73.4	71.9	64.3
	2	1.6	1.6	1.6	4.2	10.5	1.8	2.1	2.1	4.9	24.2	2.6	2.4	2.6	4.1	8.9
	3	21.5	16.2	20.4	17.8	4.7	19.2	15.5	17.4	16.2	3.1	12.6	10.2	11.5	10.8	3.0
	4	11.5	16.8	12.6	15.2	33.0	12.5	16.2	14.3	15.5	31.5	9.5	11.9	10.8	11.5	21.9
	5	3.7	3.7	3.7	3.7	3.7	3.4	3.4	3.4	3.7	3.4	1.3	1.3	1.3	1.3	1.3
	6	0.5	0.5	0.5	0.5	0.5	1.8	1.8	1.8	1.5	1.5	0.4	0.4	0.4	0.4	0.6
Den = 40	1	63.5	63.5	63.5	63.5	57.9	68.4	68.4	68.4	68.4	66.5	72.3	72.1	70.7	70.1	68.4
	2	2.6	2.1	2.1	1.7	6.9	24.6	24.6	24.6	24.6	3.8	20.3	20.5	21.9	22.3	23.4
	3	30.0	30.5	30.5	30.9	30.5	1.6	1.6	1.6	1.6	24.3	2.0	2.0	2.0	2.1	2.3
	4	0.9	0.9	0.9	0.9	1.7	2.2	2.2	2.2	2.2	2.2	2.7	2.7	2.7	2.7	2.9
	5	2.1	2.1	2.1	2.1	2.1	2.6	2.6	2.6	2.6	2.6	2.0	2.0	2.0	2.0	1.9
	6	0.9	0.9	0.9	0.9	0.9	0.6	0.6	0.6	0.6	0.6	0.8	0.8	0.8	0.8	1.0

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